DRONES

5

CHAPTER WISE INDEX - DRONES (SD/SW)

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DRONES

CHAPTER D I : EVOLUTION AND TYPES OF DRONES

"A drone, in technological terms, is an unmanned aircraft. … Essentially, a drone is a flying robot that can be remotely controlled or fly autonomously through software-controlled flight plans in their embedded systems, working in conjunction with onboard sensors and GPS"



TEACHING INSTRUCTIONS

Period	:	One (01)
Type	:	Lecture and Presentation
Year	:	1st Year SD/SW
Conducting Officer	:	Officer/ Trained PI/ AMI/ CGI
Training Aids	:	Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, and Models.

Time Plan

➢ Introduction	:	05 Min
➢ History of Drones	:	10 Min
➢ Classification of Drones	:	20 Min
➢ Conclusion	:	05 Min

INTRODUCTION

1. A drone, in technological terms, is an unmanned aircraft. Essentially, its a flying robot that can be remotely controlled or fly autonomously through software-controlled flight plans in their embedded systems, working in conjunction with on-board sensors and GPS. Drones can be rightly termed as “eye in the sky”.

2. Drones have revolutionized various industries by providing a versatile platform for a wide range of applications. These aircrafts are equipped with advanced technologies, such as high-resolution cameras, thermal sensors, and precise navigation systems, allowing them to perform tasks like aerial photography, surveillance, mapping, and search-and-rescue operations.

PREVIEW

The lecture will be conducted in the following parts:-

- Part I : Evolution of Drones
- Part II : Classification of Drones

LEARNING OBJECTIVES

- To understand the history and evolution of drones
- To learn and understand the categorisation of unmanned aircraft systems.
- Understanding the classification of unmanned aircraft systems

PART I: EVOLUTION OF DRONES

3. **Early Beginnings**. The genesis and history of drones is a fascinating journey that spans centuries, evolving from early concepts to the advanced technology we see today.

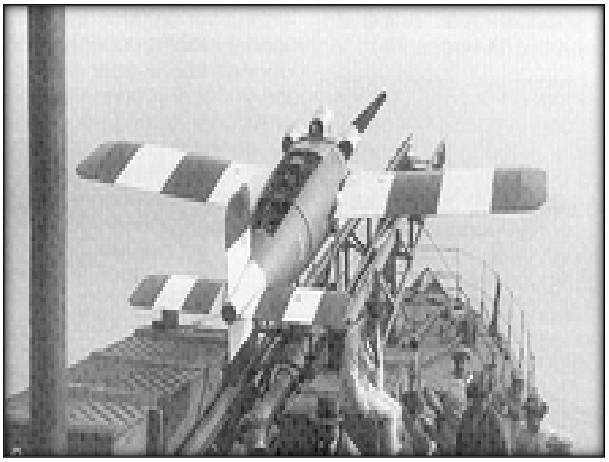
(a) **Ancient China**. The earliest use of unmanned flying devices can be traced back to ancient China, where kites were used for signaling, reconnaissance, and even lifting soldiers into the air.

(b) **Leonardo da Vinci.** In the late 15th century, da Vinci envisioned machines resembling birds and bats, capable of flight without a pilot.

4. **Military Roots**.

(a) **World War I**. The British military developed the "Aerial Target" to distract and confuse enemy defenses.

(b) **World War II**. Both the US and Germany made significant strides in drone technology. The US developed the "Radioplane OQ-2" for target practice, while Germany created the "V-1" flying bomb.

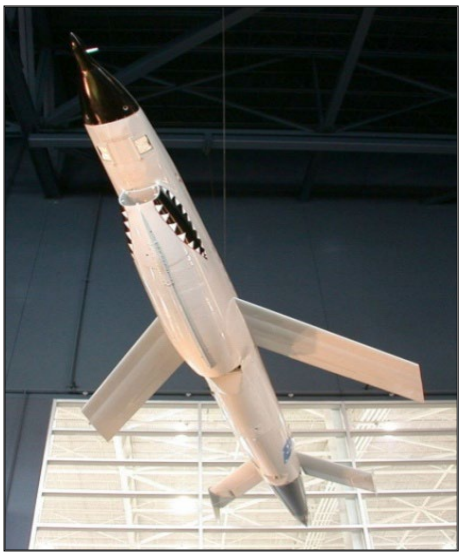




5. **Post-War Developments**.

(a) **1960s**. The US military began experimenting with remotely piloted aircraft for reconnaissance, leading to the development of the "Ryan Firebee".

(b) **1980s-1990s**. Drones started to be used for scientific research, environmental monitoring, and even entertainment.





6. **Modern Era.**

(a) **2000s**. The advent of GPS and advanced sensors led to significant advancements in drone technology.

(b) **2010 to Present Era**. Drones have become indispensable tools in various fields, including agriculture, delivery services, infrastructure inspection, emergency response and even a medium to attack enemy country through aerial route.

7. **Key Milestones**. Drones have come a long way from their early military origins to becoming versatile tools used in everyday life. Their evolution continues as technology advances, opening up new possibilities and applications. Some important milestones are as under:-

(a) **1849**. Austrian soldiers attacked the city of Venice with unmanned balloons filled with explosives.

(b) **1917.** The British "Aerial Target" was the first successful remote-controlled aircraft.

(c) **1935.** Reginald Denny developed the first civilian remotely piloted vehicle.

(d) **1960s**. The "Ryan Firebee" became one of the first successful remotely piloted drones.

8. Drones have come a long way from their early military origins to becoming versatile tools used in everyday life. Their evolution continues as technology advances, opening up new possibilities and applications.

Glossary of Terms.

9. **Drone**. It means an unmanned aircraft system which can be remotely controlled or fly autonomously through software-controlled flight plans in their embedded systems, working in conjunction with on-board sensors and GPS.

10. **Drone Acknowledgement Number**. It denotes the unique number issued by the digital sky platform under the voluntary disclosure scheme for unmanned aircraft systems in India.

11. **Geo-fencing**. It means restricting the movement of unmanned aircraft system within a defined airspace.

12. **Digital Sky Platform**. It refers to the online platform hosted by the Directorate General of Civil Aviation for various activities related to the management of unmanned aircraft system activities in India.

INTERESTING FACT

- The global drone market size was estimated at USD 64.32 billion in 2023 and is projected to grow at a Compounded annual growth rate of 14.5% from 2024 to 2030, owing to various factors, such as advances in technology, broadening application portfolio across various industries, and plummeting costs of drone technology

HIGHER ORDER THINKING SKILLS (HOTS)

- **Historical Impact.** Analyse the historical events that led to the development of drone technology. How did the needs and innovations of wartime contribute to the evolution of drones?
- **Technological Evolution**. Examine the key technological advancements that made the early development of drones possible. How did breakthroughs in radio control and material science play a role?
- **Comparative Analysis.** Compare the initial uses of drones in military contexts to their modern-day applications. How have the core principles remained the same or evolved over time?

13. **Interesting Timelines**.



Winston Churchill, David Margesson and others wait to watch the launch of de Havilland Queen Bee target drone, 06 June 1941.

US Army commissioned the Kettering Aerial Torpedo - the "Bug".

The "Bug" after a predetermined length go time, an electrical circuit used to shut off the engine and the wings were released, causing the "Bug" to plunge to earth - where its 180 lbs of explosive detonated on impact.





The next phase went from the swamps of Vietnam, to the remote hills and caves of Pakistan and Afghanistan.

The Predator air vehicle is 8.2m (27ft) long and has a 14.9m (49ft) wingspan. The system operates at an altitude of 7,620m (25,000ft) and has a range of around 740 Km.

"The drone revolution will change the way we live, work, and play." - Chris Anderson

PART II : CLASSIFICATION OF DRONES

14. Drones come in various types, each designed for specific applications and functions. They are categorized and classified in several ways.

Based on Design

15. **Multirotor Drones**.

(a) **Quadcopters**. Four rotors, used for aerial photography, surveillance, and recreational purposes.

(b) **Hexacopters**. Six rotors, offering more stability and payload capacity.

(c) **Octocopters**. Eight rotors, used for heavy-lift operations and high-end cinematography.

16. **Fixed-Wing Drones**.

(a) Similar to traditional aeroplanes, used for long-distance missions and high-endurance tasks.

(b) Commonly used in mapping, surveying, and agricultural monitoring.

17. **Hybrid VTOL Drones**.

(a) Capable of Vertical Take-off and Landing (VTOL) and efficient forward flight.

(b) Combine features of both multirotor and fixed-wing drones.

By Operational Use

18. **Model Remotely Piloted Aircraft System (MRPAS)**. Used for educational and research purposes without payload.

19. **Remotely Piloted Aircraft System (RPAS)**. Piloted from a remote pilot station for various purposes, including surveillance and deliveries.

20. **Autonomous Unmanned Aircraft System (AUAS)**. Operates autonomously without pilot intervention, often used in complex missions.

Based on Size and Weight

21. **The Unmanned Aircraft System (UAS)**. Based on the maximum all-up weight including payload, is classified as given in succeeding paragraphs.

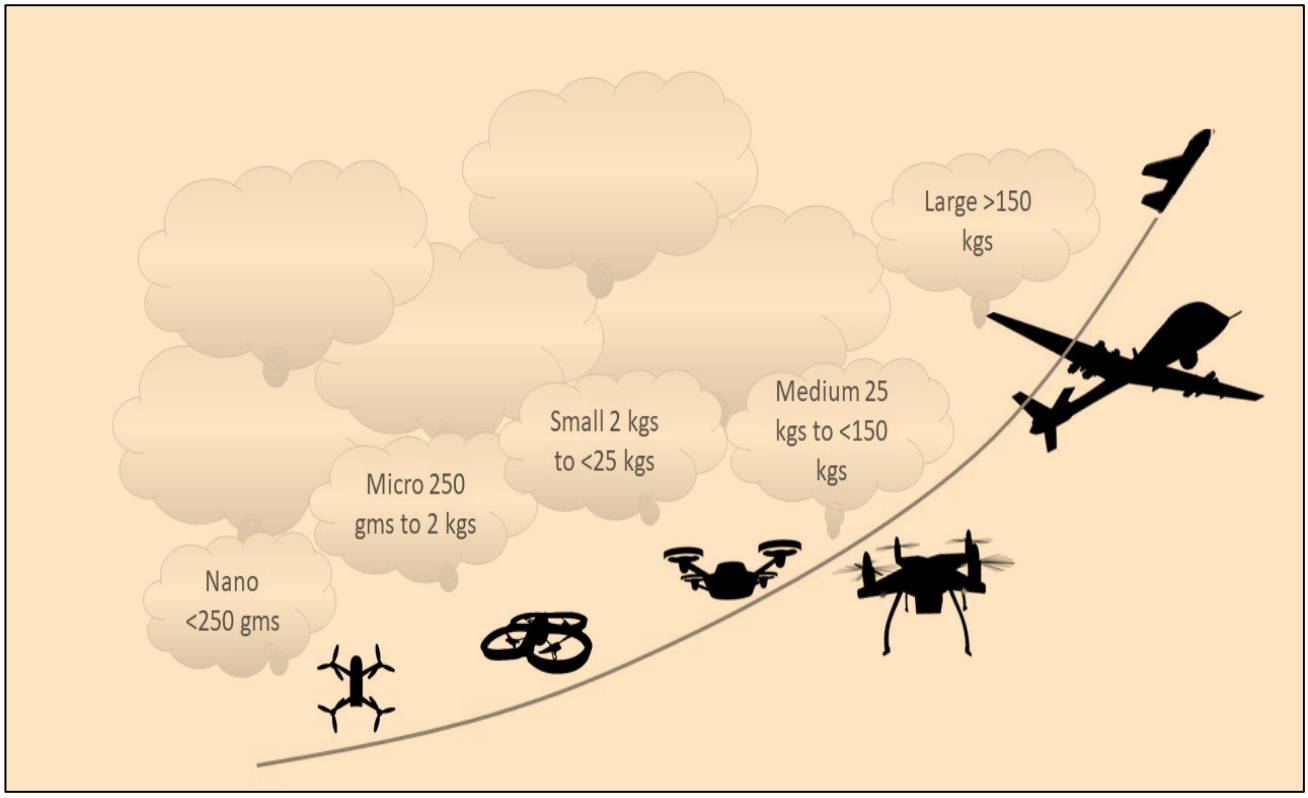
22. **Nano UAS**. Very small, often used for indoor flight and educational purposes, weighing less than or equal to 250 grams.

23. **Micro UAS**. Slightly larger than Nano drones, used for hobbyist activities and short-range surveillance, weighing more than 250 grams, but less than or equal to 2 kg.

24. **Small UAS**. Compact drones for consumer use, often used for photography and racing, weighing more than 2 kg but less than or equal to 25 kg.

25. **Medium UAS**. Used in commercial applications such as delivery and industrial inspections, weighing more than 25 kg, but less than or equal to 150 kg.

26. **Large UAS**. Heavy-duty drones used in military, cargo transport, and long-endurance surveillance missions, weighing more than 150 kg.



Based on Usage

27. **Commercial Drones**. Used in agriculture, delivery services, infrastructure inspection, and filmmaking.

28. **Military Drones**. Used for reconnaissance, combat operations, and surveillance.

29. **Recreational Drones**. Used by hobbyists for fun, aerial photography, and racing.

30. **Industrial Drones**. Used in construction, mining, and energy sectors for inspection and monitoring.

31. **Scientific Drones**. Used in research, environmental monitoring, and wildlife tracking.

Scope of Drone Operations

32. These categories are critical in understanding the scope and limitations of drone operations, influencing both technological development and regulatory frameworks. An overview of **VLOS (Visual Line of Sight)** and **BVLOS (Beyond Visual Line of Sight)** in drone operations is given in succeeding paragraphs.

33. **Visual Line of Sight (VLOS).**

(a) **Definition**. The drone is always flown within the pilot's visual line of sight.

(b) **Visibility**. The pilot must be able to see the drone without any visual aids (e.g., binoculars or FPV goggles).

(c) **Safety**. The pilot can monitor the airspace for potential hazards and maintain control of the drone.

(d) **Distance**. Typically, the maximum visibility is around 500 meters (1640 feet), but this can vary based on factors like drone size, environment, and weather conditions.

34. **Beyond Visual Line of Sight (BVLOS)**.

(a) **Definition**. The drone is flown beyond the pilot's visual line of sight.

(b) **Visibility**. The pilot does not have direct visual contact with the drone.

(c) **Safety**. Requires advanced planning, reliable communication systems, and often additional safety measures.

(d) **Applications**. Used for long-range missions, such as infrastructure inspection, delivery services, and large-scale agricultural monitoring.

35. **Key Differences.**

(a) **Control.** In VLOS, the pilot directly controls the drone. In BVLOS, the drone may operate autonomously or with remote control.

(b) **Regulations.** BVLOS operations typically require special permissions and certifications due to the increased complexity and safety considerations.

36. BVLOS operations hold great potential for expanding the capabilities of drones, but they also come with additional challenges and regulatory requirements.

HIGHER ORDER THINKING SKILLS (HOTS)

- **Interdisciplinary Integration.** Explore how the different fields of science and engineering contributed to the initial development of drones. What interdisciplinary collaborations were crucial to early drone innovation?
- **Technological Challenges**. Identify the primary challenges faced during the early development of drones. How were the challenges overcome, and what lessons were learnt that apply to current drone technology?
- **Predictive Analysis**. Based on the historical development of drones, predict the future trends in drone technology. How might the foundational principles of early drones influence future innovations?
- **Global Influence**. Assess the global influence of early drone technology on international military strategy and policy. How did the development of drones global dynamics and power structures?

CONCLUSION

37. Drones, once a futuristic concept, have rapidly evolved into versatile tools with a wide range of applications. From their humble beginnings as military targets to their current role in various industries, drones have transformed the way we perceive and interact with the world.

38. **Key Takeaways from the Evolution of Drones.**

(a) **Technological Advancements**. The integration of GPS, advanced sensors, and autonomous flight capabilities has significantly enhanced drone performance and functionality.

(b) **Diverse Applications**. Drones are used in sectors such as agriculture, logistics, construction, and entertainment, offering innovative solutions to complex challenges.

(c) **Regulatory Framework**. The development of comprehensive regulations is crucial to ensure the safe and responsible use of drones.

(d) **Future Potential**. As technology continues to advance, drones are poised to revolutionize industries and redefine the possibilities of aerial operations.

39. By understanding the history, types, and applications of drones, we can appreciate their transformative impact and anticipate their future potential.

SUMMARY

- The concept of drones dates back to 1849.
- 1915-1920 saw a giant leap forward in drone technology.
- The first pilotless aircraft was developed in 1916.
- In the 1930s, the U.S. Navy began experimenting with radio-controlled aircraft.
- " Drone Acknowledgement Number" means the unique number issued by the digital sky platform.
- "Digital Sky Platform" means the online platform hosted by the DGCA.
- The UAS is categorised as Aeroplane, Rotorcraft and Hybrid.
- UAS is based on the maximum all-up weight.
- **Multirotor Drones**. Quadcopters, Hexacopter, Octocopters.
- **Fixed-Wing Drones**. Airplane-like drones.

- **Hybrid VTOL Drones**. Combines multirotor and fixed-wing features.
- **MRPAS**. Model Remotely Piloted Aircraft System
- **RPAS**. Remotely Piloted Aircraft System
- **AUAS**. Autonomous Unmanned Aircraft System
- **VLOS (Visual Line of Sight)**. Pilot maintains visual contact.
- **BVLOS (Beyond Visual Line of Sight)**. Drone operates autonomously or with remote control.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What does the term "Drone" refer to?

(a) A manned aircraft system

(b) A remote control vehicle

(c) An unmanned aircraft system

(d) A satellite-based communication system

Q2. What is a "Drone acknowledgement number"?

(a) The registration number issued for manned aircrafts

(b) The unique number issued by the digital sky platform for unmanned aircraft systems

(c) A tracking number for drone deliveries

(d) A pilot's certification number

Q3. Which of the following is NOT a person subject to the Drone Rules?

(a) A person owning a drone

(b) A person operating a drone in India

(c) A person selling manned aircraft

(d) A person maintaining a drone

Q4. To which unmanned aircraft systems do the Aircraft Rules, 1937 still apply?

(a) Drones used by private companies

(b) Unmanned aircraft systems with a maximum all-up-weight of more than 500 kilograms

(c) All drones regardless of weight

(d) Drones operated for research purposes

Q5. Which entity operates the "Digital Sky Platform"?

(a) The Indian Navy

(b) The Ministry of Communications

(c) The Directorate General of Civil Aviation

(d) The Indian Meteorological Department

Q6. What is the primary characteristic of an "Aeroplane" in the context of unmanned aircraft systems?

(a) It is a lighter-than-air aircraft.

(b) It derives lift from fixed aerodynamic surfaces.

(c) It uses rotors for lift during flight.

(d) It relies on engine thrust only for lift.

Q7. How is a "Rotorcraft" supported in flight?

(a) By a fixed-wing surface

(b) By the reactions of air on one or more power-driven rotors on vertical axes.

(c) By gliding on air currents.

(d) By a combination of fixed-wing and rotor-based mechanisms.

Q8. Which of the following best describes a "Hybrid Unmanned Aircraft System"?

(a) It relies on fixed surfaces for vertical and horizontal flight.

(b) It is supported by rotating airfoils during vertical flight only.

(c) It is capable of vertical take-off and uses non-rotating airfoils during horizontal flight.

(d) It has no capability for vertical take-off and landing.

Q9. Which of the following is NOT a sub-category of unmanned aircraft systems?

(a) Remotely piloted aircraft system

(b) Autonomous unmanned aircraft system

(c) Semi-autonomous manned aircraft system

(d) Model remotely piloted aircraft system

Q10. What is the maximum weight limit for a Nano UAS?

(a) 250 grams

(b) 2 kilograms

(c) 25 kilograms

(d) 150 kilograms

Q11. Which classification includes unmanned aircraft weighing more than 250 grams but less than or equal to 2 kilograms?

(a) Nano UAS

(b) Micro UAS

(c) Small UAS

(d) Medium UAS

Q12. Which classification applies to unmanned aircraft weighing more than 25 kilograms but less than or equal to 150 kilograms?

(a) Nano UAS

(b) Small UAS

(c) Medium UAS

(d) Large UAS

Q13. Which category of unmanned aircraft includes vehicles weighing more than 150 kilograms?

(a) Small UAS

(b) Medium UAS

(c) Large UAS

(d) Micro UAS

Q14. What is the primary function of a "Remotely piloted aircraft system"?

(a) It operates autonomously with no human intervention.

(b) It is controlled by a human operator using a remote system.

(c) It is used for recreational purposes only.

(d) It is controlled by on-board pilots.

Q15. Which of the following unmanned aircraft systems can perform vertical take-off and landing?

(a) Aeroplane

(b) Rotorcraft

(c) Hybrid unmanned aircraft system

(d) Nano UAS

Fill in the Blanks

Q1. The _____ is an online platform for managing unmanned aircraft systems activities in India.

Q2. The _____ Rules, 1937 do not apply to unmanned aircraft systems except those with a maximum all-up-weight of more than 500 kilograms.

Q3. An unmanned aircraft system used by the naval, military, or air forces of India is _____ from the Drone Rules.

Q4. A "Drone acknowledgement number" is issued by the _____ under the voluntary disclosure scheme.

Q5. The Drone Rules apply to all unmanned aircraft systems that are being operated in or over _____.

Short Answer Questions

Q1. Which year saw the development of the first pilotless aircraft?

Q2. Which organisation experimented with radio-controlled aircraft in the 1930s?

Q3. What does the term "Drone Acknowledgement Number" refer to?

Q4. What is the "Digital Sky Platform"?

Q5. How are UAS categorised based on their maximum all-up weight?

Q6. Name three types of multirotor drones.

Q7. What type of drones are similar to traditional aeroplanes?

Long Answer Questions

Q1. Discuss the evolution of drone technology from its inception in 1849 to the early 20th century.

Q2. Explain the significance of the development of the first pilotless aircraft in 1916 and its impact on subsequent drone technology.

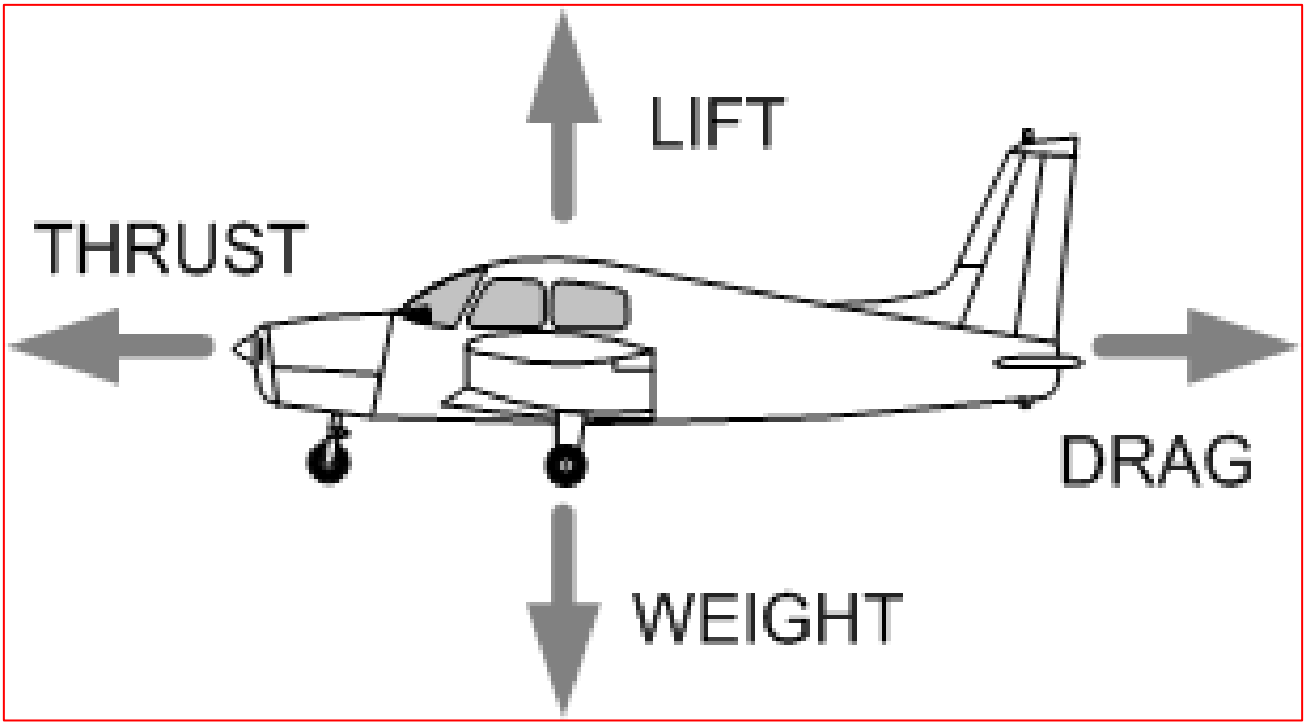
Q3. Analyse the role and importance of the Digital Sky Platform and the Drone Acknowledgement Number in the regulation of drone operations in India.

Q4. Compare and contrast the different types of UAS categorised by their design and operational use, including aeroplanes, rotorcraft, and hybrids.

Q5. What does VLOS stand for in drone operations? Describe how VLOS systems operate.

DRONES

CHAPTER D II : BASIC PRINCIPLES OF FLIGHT



TEACHING INSTRUCTIONS

Period : Two (02)

Type : Lecture and Presentation

Year : 2nd Year SD/SW

Conducting Officer : Officer/ Trained PI/AMI/CGI

Training Aids : Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, and Models.

Time Plan

- Introduction : 05 min
- Fundamentals of Flight : 20 min
- Control Surfaces of an Aircraft : 15 min
- Flight Proper : 20 min
- Manoeuvres, Phases and Circuit Pattern : 10 min
- Q & A Session & Conclusion : 10 min

INTRODUCTION

1. Aerodynamics plays a crucial role in designing aircraft to maximise efficiency and safety. Understanding how air interacts with the aircraft's surfaces helps engineers improve performance and stability. These principles are fundamental to the science of aviation, ensuring that aircraft can fly safely and efficiently through the balance of these forces and controls.

2. The basic principles of flight for drones are similar to those of fixed-wing and rotary, aircrafts but they are more specialized due to their unique design and technology. To stay in the air, like aircrafts, drones also rely on four main forces of **lift**, **weight**, **thrust**, and **drag**. Drone technology is woven around the interplay of these forces.

PREVIEW

The lecture will be conducted in the following parts:-

- Part I : Fundamentals of Flight
- Part II : Takeoff, Flight, and Landings
- Part III : Manoeuvers and Turns

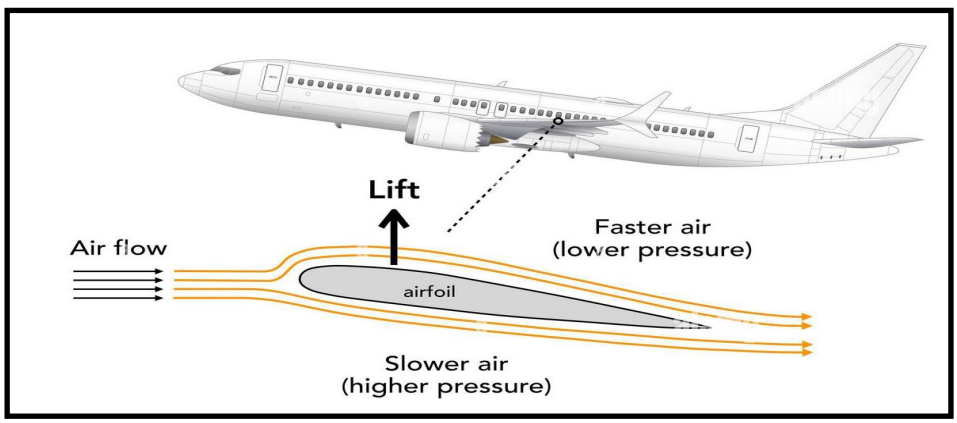
LEARNING OBJECTIVES

- To understand the basic principle of flight.
- To learn about primary controls & movements.
- Understanding the aircraft manoeuvers and circuit pattern.

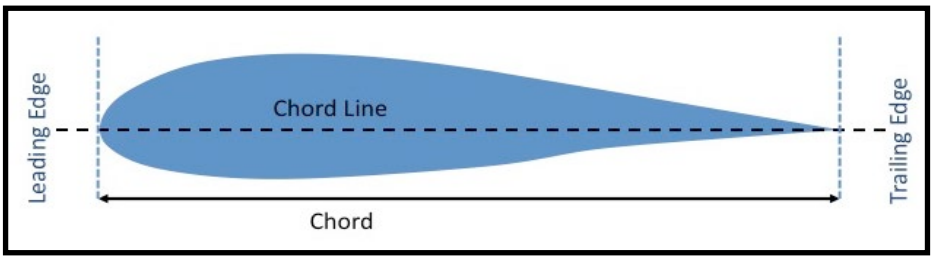
PART I : FUNDAMENTALS OF FLIGHT

Glossary of Terms

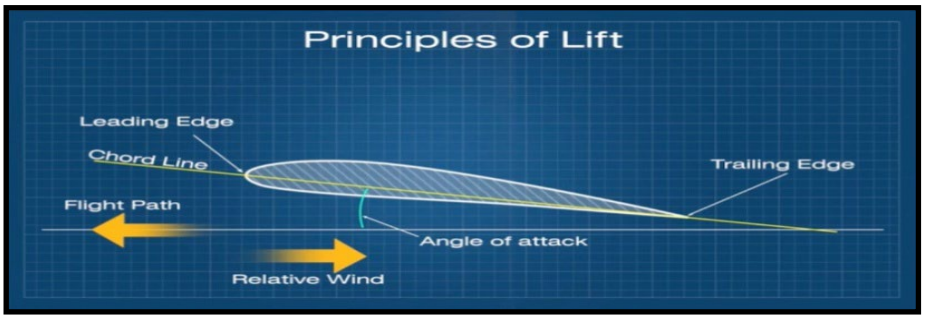
3. **Airfoil**. An **Airfoil** is a streamlined shape designed to produce more lift than drag when moving through a fluid.



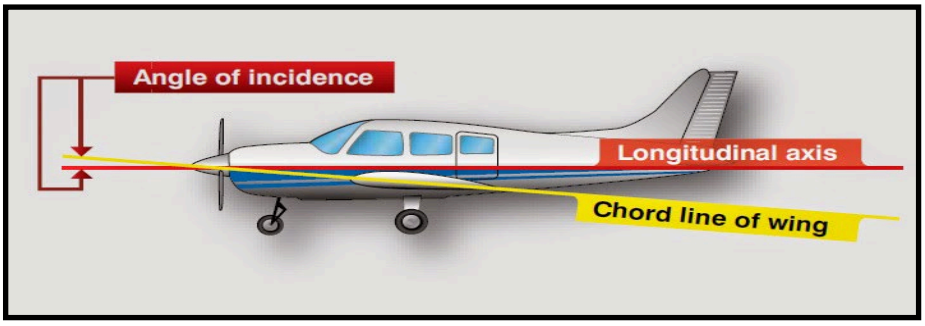
4. **Chord Line**. It is a line joining the centres of curvature of leading and trailing edges of an airfoil section.



5. **Angle of Attack (AoA)**. It is an angle between the chord line of the airfoil and the relative airflow.



6. **Angle of Incidence**. The angle between the chord line and the longitudinal axis of the aircraft is called Angle of Incidence.

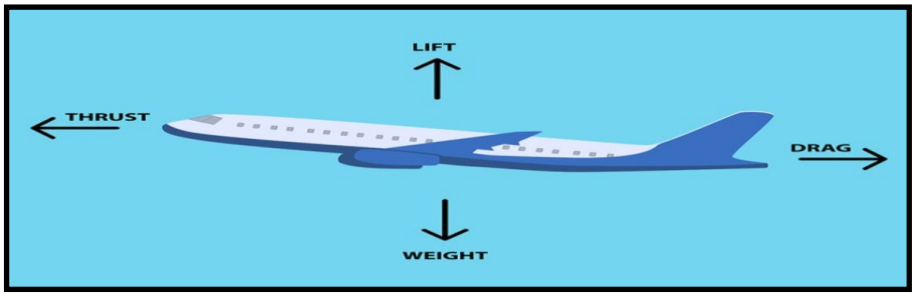


Aerodynamic Principles

7. The aerodynamic principles of flight form the foundation of how objects like airplanes, helicopters, and drones achieve and maintain flight.

Forces Acting on an Aircraft

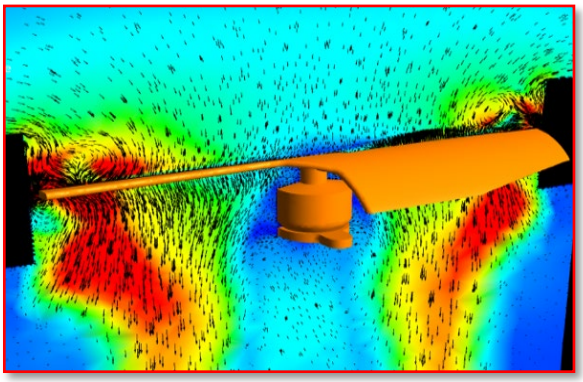
8. The aerodynamic principles revolve around four key forces which are lift, drag, thrust and weight. These forces all interact together to determine an aeroplane's trajectory. Lift and weight are opposing forces, as are thrust and drag. All are equally important and must be balanced to maintain level flight.

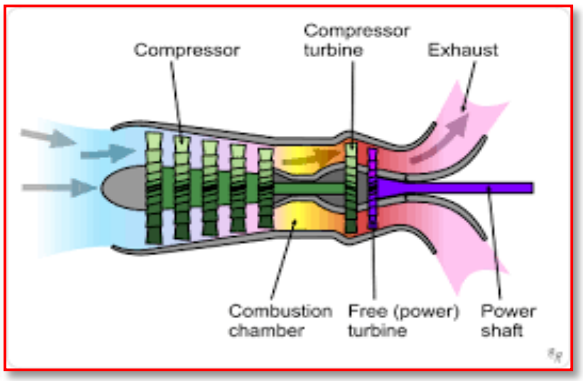


(a) **Weight**. Weight is the force of gravity pulling the object downward toward the Earth. Weight is proportional to the mass of the object and acts through its center of gravity. In flight, the force of weight is countered by the forces of lift and thrust.

(b) **Lift.** Lift is the upward force that opposes the downward pull of gravity or weight. It is a positive force generated by the movement of air over an airfoil e.g., wings or propellers because of the difference in air pressure under and above a wing. The higher air pressure beneath a wing creates lift. Volume of lift generated is affected by the shape of the wing.**To achieve flight, lift must equal or exceed weight.**

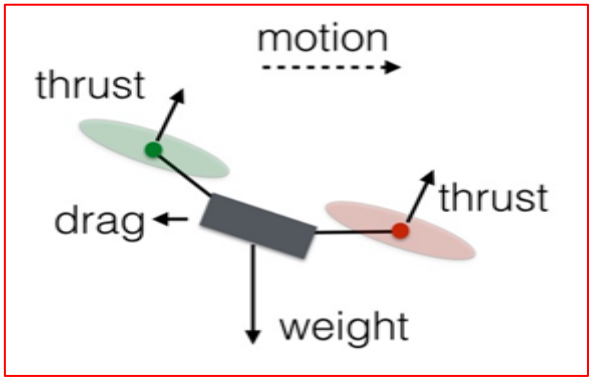
(c) **Thrust**. Thrust is the **forward force** that propels an object through the air. Thrust is typically generated by engines, propellers, or jet turbines, through some kind of propulsion system. For drones, the spinning rotors create thrust by pushing air downward and forward, propelling the drone.





(d) **Drag** is the resistance of the air to anything moving through it. Drag is the opposing force to thrust. It is caused by aerodynamic resistance as an object moves through the air. Drag is an umbrella term which can include form drag, parasite drag etc.

(e) The four forces making up the principle of flight are **lift**, **weight**, **drag**, and **thrust**. The forces all interact together to determine an airplane's trajectory. Lift and weight are opposing forces, as are thrust and drag. All are equally important, and they must be balanced to maintain a level flight.



PART II : CONTROL SURFACES AND AXIS OF AN AIRCRAFT

9. The control surfaces work in conjunction with the aircraft's axis of rotation to provide precise control over its flight path. By understanding these fundamental principles, pilots can safely and efficiently manoeuver their aircraft. An aircraft's ability to manoeuver in three dimensions is made possible by its control surfaces and axis of rotation. These elements work in tandem to allow pilots to precisely control the aircraft's flight path.

10. **Control Surfaces**.

(a) **Ailerons.** These are hinged sections on the trailing edge of each wing. By deflecting the ailerons differentially, the pilot can create a difference in lift between the wings, causing the aircraft to roll.

(b) **Elevators.** These are hinged sections on the trailing edge of the horizontal stabilizer. By deflecting the elevators up or down, the pilot can control the pitch of the aircraft.

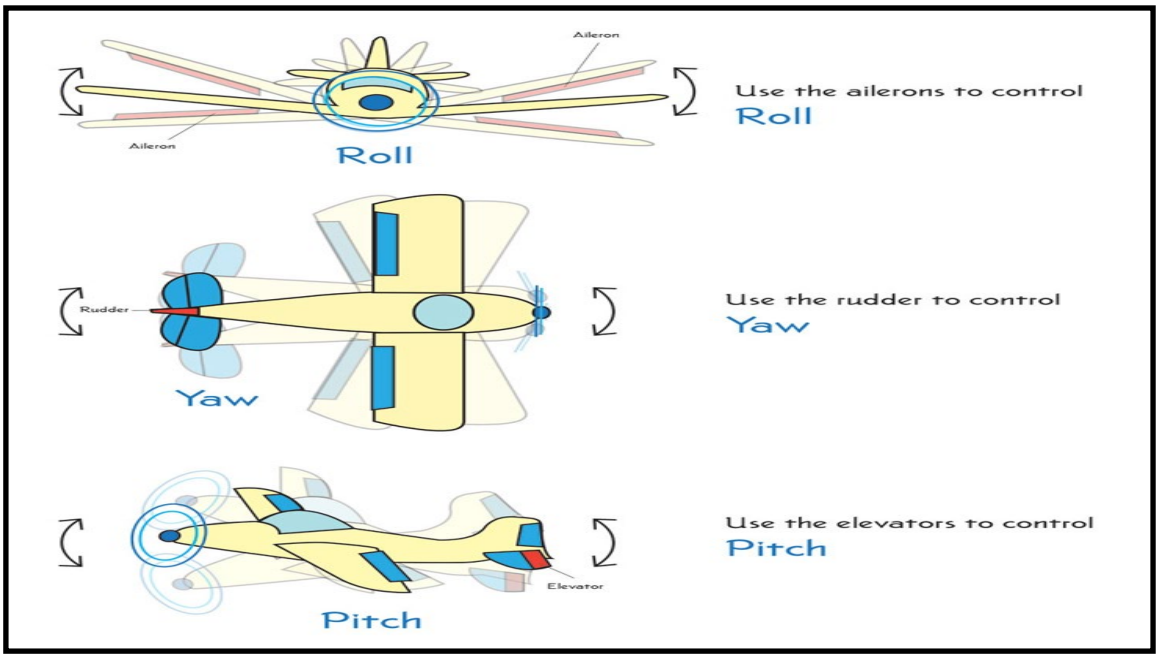
(c) **Rudder.** The rudder is a hinged section on the trailing edge of the vertical stabilizer. By deflecting the rudder left or right, the pilot can control the yaw of the aircraft.

11. **Axis of Rotation**.

(a) **Lateral Axis (Roll).** This axis runs from wingtip to wingtip. Rolling the aircraft involves tilting its wings left or right.

(b) **Longitudinal Axis (Pitch).** This axis runs from the nose to the tail of the aircraft. Pitching involves raising or lowering the nose of the aircraft.

(c) **Vertical Axis (Yaw).** This axis runs vertically through the aircraft's center of gravity. Yawing involves turning the nose of the aircraft left or right.

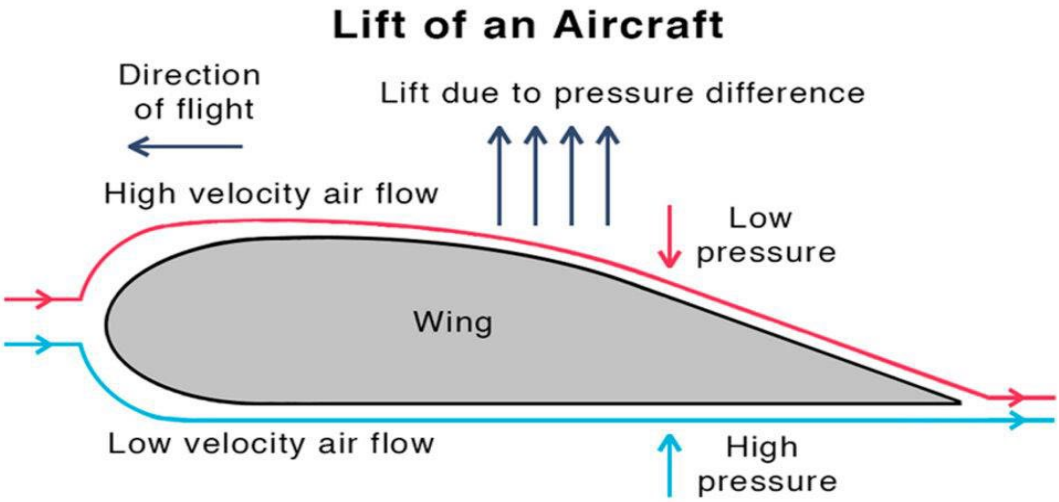


PART III : FLIGHT PROPER

12. Importance of Angle of Attack in Flight. As mentioned in glossary, Angle of Attack is the angle between the chord line of an airfoil and the relative airflow. Increasing the angle of attack increases lift up to a certain point. If the angle becomes too steep (called Critical Angle of Attack), airflow separates from the airfoil's surface, causing a stall (a sudden loss of lift). Pilots adjust the angle of attack to control lift during takeoff, cruising, and landing.

How Wings Lift an Aeroplane or a Fixed Wing Drone.

13. Aeroplane wings are shaped to make air move faster over the top of the wing. When air moves faster, the pressure of the air decreases. So, the pressure on the top of the wing is less than the pressure on the bottom of the wing. The difference in pressure creates a force on the wing that lifts the wing up into the air.



How a

Quadcopter or Drone with Four Rotors Flies.

14. A quadcopter, or drone with four rotors, flies by controlling the speed and direction of its motors to generate lift and manoeuvre in the air. Here is how it works:-

(a) **Lift Generation**. Each of the four rotors (propellers) spins at high speed to generate upward force (lift). This is what keeps the quadcopter in the air. The rotors are arranged in two pairs, two spin clockwise (CW) and two spin counter-clockwise (anticlockwise) (CCW). The counteracting spins cancel out any rotational torque that would cause the drone to spin uncontrollably.

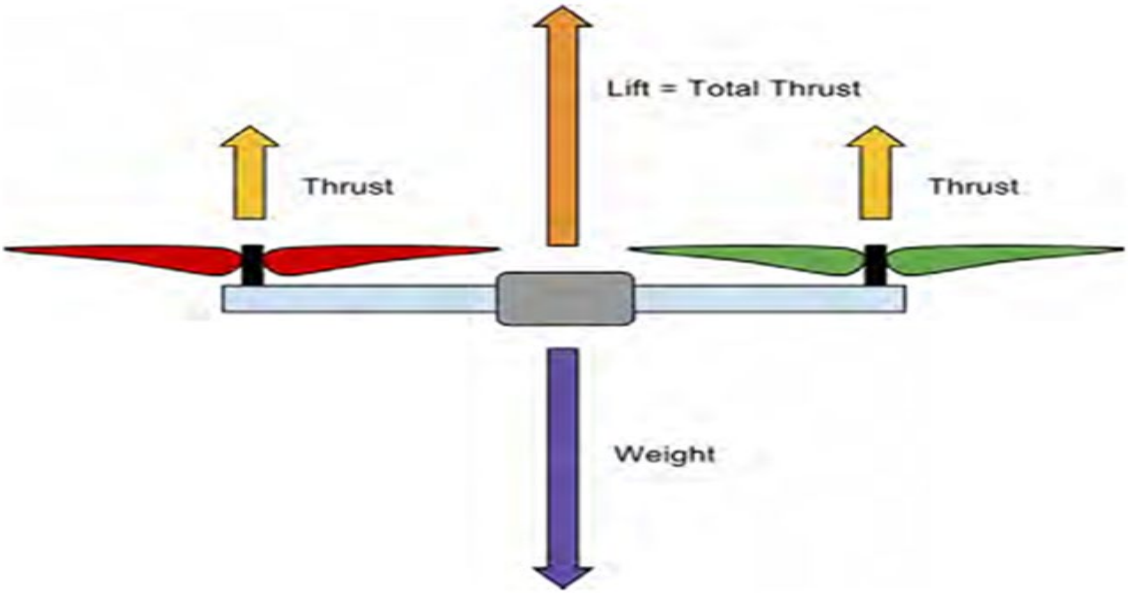
(b) **Pitch (Forward/Backward Movement)**. To pitch the quadcopter forward or backward (i.e., move it in that direction), the speed of the front or rear rotors is adjusted. Increasing the speed of the rear rotors and decreasing the speed of the front rotors causes the drone to tilt forward, pushing it in that direction. Similarly, the opposite action moves it backward.

(c) **Roll (Side-to-Side Movement)**. For roll (side-to-side movement), the left or right rotor speeds are adjusted. If the left rotors are spun faster and the right rotors slower, the quadcopter tilts to the right. The opposite applies for a roll to the left.

(d) **Yaw (Rotation or Turning)**. Yaw involves the rotation of the quadcopter around its vertical axis. This is controlled by adjusting the speeds of the clockwise and anticlockwise spinning rotors. For example, increasing the speed of the CW rotors and decreasing the speed of the CCW rotors will cause the quadcopter to rotate clockwise. This rotation allows the quadcopter to change its direction without needing a rudder like in traditional aeroplanes.

(e) **Altitude Control**. The quadcopter's altitude is controlled by the overall speed of all four rotors. When all rotors spin faster, they generate more lift, causing the quadcopter to ascend. Slowing the rotors down will reduce lift, making the quadcopter descend.

(f) **Hover**. Where fixed wing aircraft use thrust to create forward motion and then use wings to convert this forward motion into lift, but **in multi rotors, lift equals total thrust**. This only happens when the drone is level.



PART IV : MANOEUVERS, PHASES AND CIRCUIT PATTERN

15. **Basic Aircraft Manoeuvers**. In aviation, manoeuvers and turns are essential techniques pilots use to control their aircraft.

(a) **Straight and Level Flight**. Maintaining a constant altitude and heading.

(b) **Climbs**. Increasing altitude by increasing the angle of attack and/or thrust.

(c) **Descents**. Decreasing altitude by reducing thrust and/or increasing the angle of attack.

(d) **Turns**. Changing direction by banking the aircraft and adjusting the rudder.

16. **Phases of Flight**. The **phases of flight** describe the various stages an aircraft goes through from the moment it starts moving on the ground until it comes to a complete stop after landing. Understanding these phases is crucial for ensuring safety and efficiency in aviation.

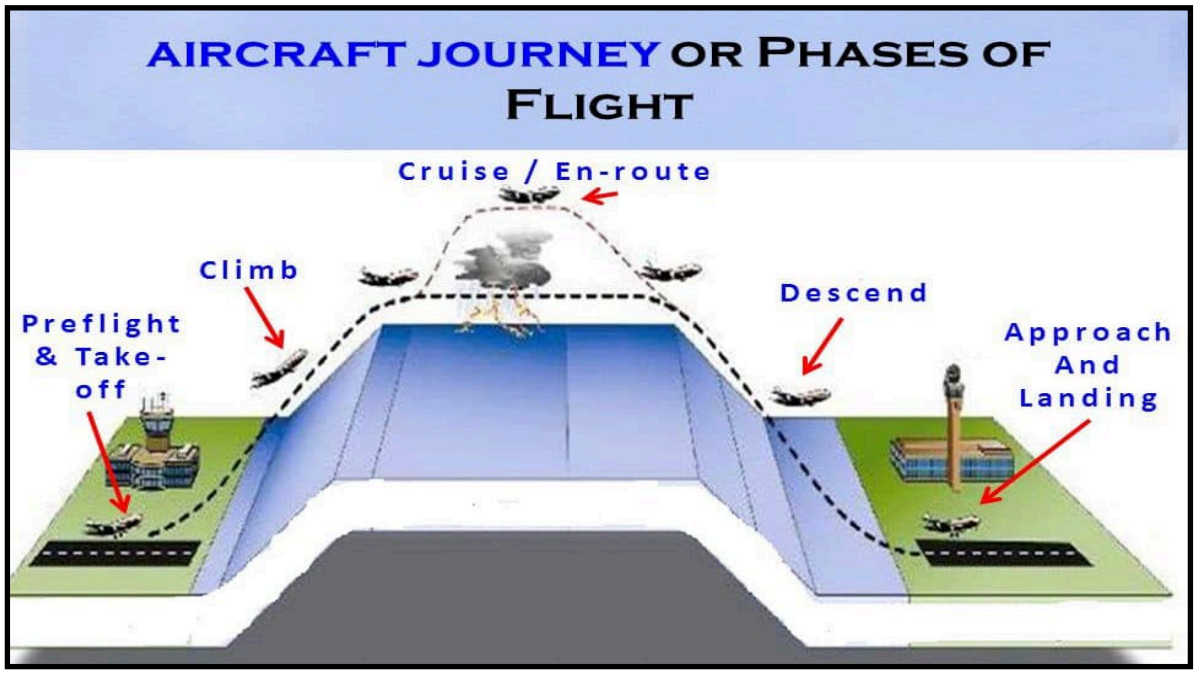
(a) **Takeoff**. When the aircraft accelerates along the runway and lifts off the ground, it is called 'takeoff'.

(b) **Climb**. When the aircraft gains altitude after takeoff, it is called 'climb'.

(c) **Cruise**. When the aircraft maintains a steady altitude and speed, it is called 'Cruise'.

(d) **Descent**. When the aircraft loses altitude before landing, it is called a 'descent'.

(e) **Landing**. When the aircraft slows down and touches down on the runway safely, it called a 'landing'.



17. Understanding these phases helps pilots, air traffic controllers, and ground crew to coordinate and ensure the safety and efficiency of flight operations. Each phase has specific procedures and protocols to maintain safety and compliance with aviation regulations. These well-defined stages not only ensure smooth operations but also enhance the overall safety.

18. **Circuit Pattern.** A circuit pattern is a standardized flight path used by pilots to practice basic flying skills and approach landing procedures. By understanding these fundamental manoeuvers, turns, and circuit patterns, pilots can safely and efficiently operate their aircraft. They typically involve the following stages -

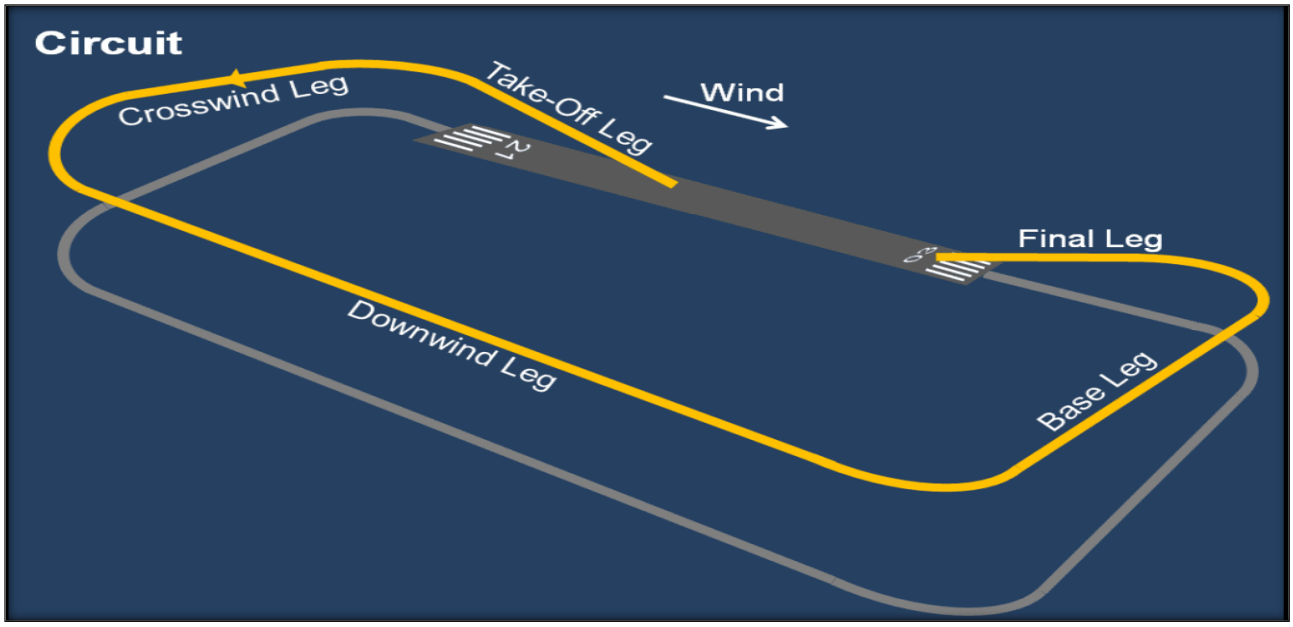
(a) **Take-off Leg**. The aircraft flies into the wind to maintain stability.

(b) **Crosswind Leg**. The aircraft turns to fly perpendicular to the wind.

(c) **Downwind Leg**. The aircraft flies perpendicular to runway and with the wind to gain speed.

(d) **Base Leg**. The aircraft turns towards the runway.

(e) **Final Approach**. The aircraft descends and aligns with the runway.



CONCLUSION

19. Understanding the fundamental principles of flight is essential for appreciating the complex interplay of forces that allow fixed-wing and rotor drones to soar through the skies. The study of air in motion and its interaction with solid surfaces, particularly the aerofoil dictates the shape of drone wings. A balance in the **four forces acting on any drone or aircraft, namely,** lift, weight, thrust, and drag is necessary for a stable flight. The **control surfaces of** ailerons, elevators and rudders enable pilots to control the aircraft's pitch, roll, and yaw. By mastering these principles, pilots can operate drones safely and efficiently, ensuring the continued advancement of drone aviation technology and its contribution to global connectivity.

SUMMARY

- A chord line is a line joining the centres of curvature of the leading and trailing edges of an aerofoil section.
- Angle of Attack is an angle between the chord line of an aerofoil and the relative airflow.
- Angle of Incidence is the angle between the chord line and the longitudinal axis of the aircraft.
- The four forces are lift, weight, drag, and thrust.

- There are three axes around which all aircraft move.
- The primary controls are the ailerons, elevator, and rudder.
- There are four fundamental basic flight manoeuvres upon which all flying tasks are based.
- Faster fluid flow leads to lower pressure.
- **Aeroplane wings** generate lift by creating pressure differences.
- The concept of drones may well date back to 1849.
- 1915-1920 saw a giant leap forward in drone technology.
- BVLOS (Beyond Visual Line of Sight): The drone operates autonomously or with remote control. BVLOS requires advanced technology and strict regulations.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Which of the following is NOT one of the four forces of flight?

(a) Lift (b) Weight

(c) Friction (d) Thrust

Q2. Which two forces are directly opposing in flight?

(a) Lift and drag (b) Thrust and weight

(c) Lift and weight (d) Drag and lift

Q3. What is the primary function of ailerons on an aircraft?

(a) To control yaw (b) To control pitch

(c) To control roll (d) To control thrust

Q4. What does the rudder control on an aircraft?

(a) Pitch (b) Roll

(c) Yaw (d) Lift

Q5. Around which axis does an aircraft pitch?

(a) Longitudinal (b) Vertical

(c) Lateral (d) Diagonal

Q6. Which axis controls the yaw of an aircraft?

(a) Longitudinal (b) Vertical

(c) Lateral (d) Diagonal

Q7. Which of the following is required for heavier-than-air flight?

(a) Drag must exceed thrust (b) Weight must exceed lift

(c) Lift must balance weight (d) Thrust must balance drag

Q8. Which aircraft phase involves the transition from moving on the ground to flying in the air?

(a) Landing (b) Takeoff

(c) Taxiing (d) Climbing

Q9. Which aircraft type does NOT require a runway for takeoff?

(a) Fixed-wing aircraft (b) Helicopters

(c) Commercial jets (d) Airliners

Q10. What is the last phase of flight where an aircraft returns to the ground?

(a) Takeoff (b) Climb

(c) Descent (d) Landing

Q11. Which manoeuvre is NOT one of the four basic flight manoeuvres?

(a) Turns (b) Climbs

(c) Spirals (d) Descents

Q12. What is the purpose of the elevator in an aircraft?

(a) To control roll (b) To control pitch

(c) To control yaw (d) To control lift

Q13. What flight manoeuvre involves increasing the altitude of the aircraft?

(a) Turn (b) Climb

(c) Descent (d) Roll

Q14. What is the key factor to manage during landing to ensure a safe descent?

(a) Pitch control (b) Lift and weight balance

(c) Thrust exceeding drag (d) Straight-and-level flight

Q15. What is the primary focus of circuit flying?

(a) Practicing aerobatics (b) Learning take-offs, turns, and landings

(c) Flying in adverse weather (d) Mastering high-speed flight

True or False

Q1. Lift and weight must be equal to maintain level flight.

Q2. The rudder controls the roll of the aircraft.

Q3. Thrust is the force that opposes drag.

Q4. The elevator controls the yaw of an aircraft.

Q5. An aircraft pitches around its longitudinal axis.

Q6. Landing requires balancing lift and weight for a safe descent.

Q7. Heavier-than-air flight requires drag to exceed thrust.

Q8. The vertical axis of an aircraft is responsible for yaw movements.

Q9. Ailerons are primarily used to control the pitch of the aircraft.

Q10. Takeoff is the phase of flight where an aircraft transitions from ground movement to air travel.

Q11. Helicopters and VTOL aircraft do not need a runway for takeoff.

Q12. During straight-and-level flight, all forces (lift, weight, drag, and thrust) are balanced.

Q13. Turns, climbs, descents, and rolls are the four fundamental flight manoeuvres.

Q14. Circuit flying helps develop a pilot's ability to separate and manage air traffic.

Q15. A climb is when an aircraft loses altitude.

Short Answer Questions

Q1. What is the primary function of an aerofoil?

Q2. Define the chord line of an aerofoil section.

Q3. What is the angle of attack in relation to an aerofoil?

Q4. What is the angle of incidence in aviation?

Q5. Name the four forces acting on an aircraft.

Q6. How many axes do aircraft move around?

Q7. What are the primary flight controls of an aircraft?

Q8. Name the four fundamental basic flight manoeuvres.

Q9. What happens to pressure when fluid flow increases?

Q10. How do aeroplane wings generate lift?

Long Answer Questions

Q1. Discuss how the angle of attack affects the lift generated by an aerofoil.

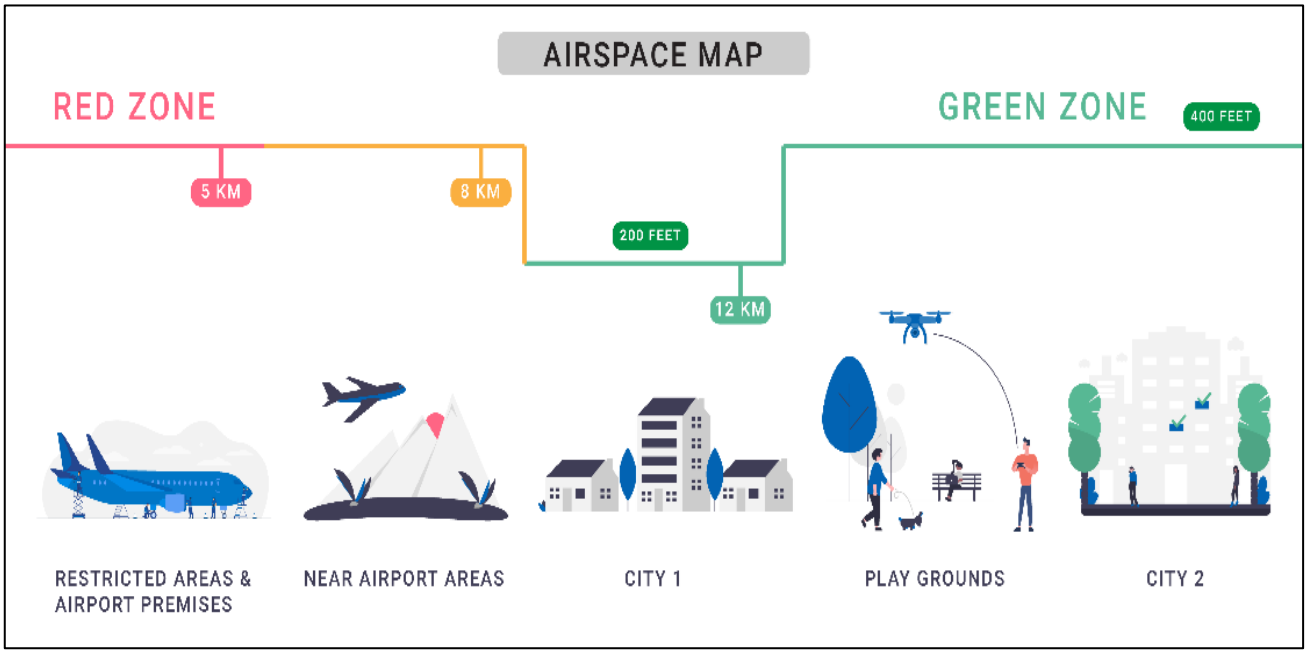
Q2. Describe the four fundamental forces acting on an aircraft and their interactions during flight. Explain the three axes of movement for aircraft and how they affect flight dynamics.

Q3. Discuss the functions of the primary flight controls: ailerons, elevator, and rudder.

Q4. Outline the four fundamental basic flight manoeuvres and their importance in aviation training.

DRONES

CHAPTER D III : DGCA RULES ON DRONES



TEACHING INSTRUCTIONS

Periods	:	Two (02)
Type	:	Lecture and Presentation
Year	:	2nd Year SD/SW
Conducting Officer	:	Officer/ Trained PI/ AMI/CGI
Training Aids	:	Class Room, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, and Models.

Time Plan

➢ Introduction	:	05 min
➢ Registration of UAS	:	10 min
➢ RPC, Extension & Validity	:	25 min
➢ RPTO	:	20 min
➢ Safety Guidelines by DGCA	:	15 min
➢ Q & A Session and Conclusion	:	05 min

INTRODUCTION

1. The Directorate General of Civil Aviation (DGCA) is the national authority responsible for regulating the use of drones in India. To ensure safety and efficiency, the DGCA has set specific rules that govern how drones should be operated. These rules cover a range of aspects, such as registration, pilot training, flight permissions, and no-fly zones, all aimed at preventing accidents and ensuring drones are used responsibly.

2. The regulations also focus on ensuring that drones do not interfere with manned aircraft and that they operate within legal and ethical boundaries. For anyone wishing to operate a drone, following DGCA guidelines is crucial to ensure safe and legal flights.

PREVIEW

The lecture will be conducted in the following parts:-

- Part I : Registration of UAS
- Part II : RPC, Extension & Validity
- Part III : RPTO
- Part IV : Safety Guidelines by DGCA

LEARNING OBJECTIVES

- To understand DGCA rules before flying Drones.
- To learn about Remote Pilot Certification and training Organisation.
- Understanding to Safety guidelines by DGCA

PART I : REGISTRATION OF UNMANNED AIRCRAFT SYSTEM

3. **General**. No person shall operate an unmanned aircraft system without first registering it on the digital sky platform and obtaining a unique identification number, unless exempted from the requirement of a unique identification number under these rules.

4. **Application and Procedure for Registration**. Any person who intends to register and obtain a unique identification number for his unmanned aircraft system shall make an application in Form D-2 on the Digital Sky platform along with the fee.

5. **Transfer of Unmanned Aircraft Systems**. A person may transfer an unmanned aircraft system to another person by way of sale, lease, gift, or any other mode after providing requisite details of the transferor, transferee, and unique identification number of the unmanned aircraft system in Form D-3 on the Digital Sky platform, along with the fee.

6. **De-Registration of Unmanned Aircraft Systems**. Where an unmanned aircraft system registered in a person's name is either permanently lost or permanently damaged, they shall, on arriving at a reasonable conclusion that it is so lost or damaged, apply for de-registration of such unmanned aircraft system by submitting an application in Form D-3 on the Digital Sky platform along with the fee.

PART II : RPL, EXCEPTION AND VAILIDITY

7. **General**. No individual other than a holder of a valid Remote Pilot License(RPL) enlisted on the Digital Sky platform shall operate an unmanned aircraft system.

8. **Classification**. A remote pilot licence shall specifically mention the category, sub-category and classification of the unmanned aircraft system or a combination of these, for which it is issued.

9. **Eligibility**. The following are the criteria under which an individual shall be eligible to obtain a remote pilot license:-

(a) If they are not less than eighteen years of age and not more than sixty-five years of age.

(b) If an individual has successfully passed the class tenth examination or its equivalent from a recognised board.

(c) If the individual has successfully completed such training as may be specified by the Director General of Govt Aviation from any authorised remote pilot training organisation.

10. **Procedure for Obtaining a Remote Pilot Licence (RPL)**.

(a) Any individual who desires to obtain a remote pilot license for any category shall complete the training specified by the DGCA and pass the tests conducted by the authorised remote pilot training organisation.

(b) Within seven days of successfully completing the training and passing the required test under an authorised Remote Pilot Training Organisation (RPTO), an application for a remote pilot license in Form D-4 shall be submitted on the Digital Sky platform along with the applicable fee. The remote pilot certificate shall then be issued through the Digital Sky platform.

11. **Validity of license**. A remote pilot license shall be valid only if it is enlisted on the Digital Sky platform unless suspended or cancelled, and is remaining valid for a period of ten years.

12. **Exemption from Obtaining a License**. No remote pilot license shall be required for:-

(a) Operating a Nano unmanned aircraft system.

(b) Operating a micro unmanned aircraft system for non-commercial purposes.

PART III : REMOTE PILOT TRAINING ORGANISATION (RPTO)

13. **General**. No person other than an authorised remote pilot training organisation shall impart training to an individual seeking a remote pilot license.

14. **Eligibility**. No remote pilot training organisation shall be authorised to impart training unless it meets the eligibility criteria as may be specified by the Director General of Civil Aviation.

15. **Procedure for Obtaining Authorisation**. Any person who intends to obtain the authorisation to establish a remote pilot training organisation shall submit an application to the Director General of Civil Aviation in Form D-5 on the Digital Sky platform, along with the fees.

16. **Validity**. An authorisation to establish a remote pilot training organisation shall, unless suspended or cancelled, remain valid for a period of ten years.

PART IV : DIGITAL SKY PLATFORM

17. Digital Sky Platform(DSP) means the online platform hosted by the Directorate General of Civil Aviation (DGCA) for various activities related to the management of unmanned aircraft system activities in India. Digital Sky is a web portal for registration, certification, authorisation, and flight planning of Unmanned Aircraft Systems (UAS) in India. It provides various services such as type certificate, unique identification number, transfer, remote pilot certificate, training organisation, and airspace map.

18. **Features of the Digital Sky Platform**.

(a) **Drone Registration**. Drones must be registered on the platform before they can be operated legally.

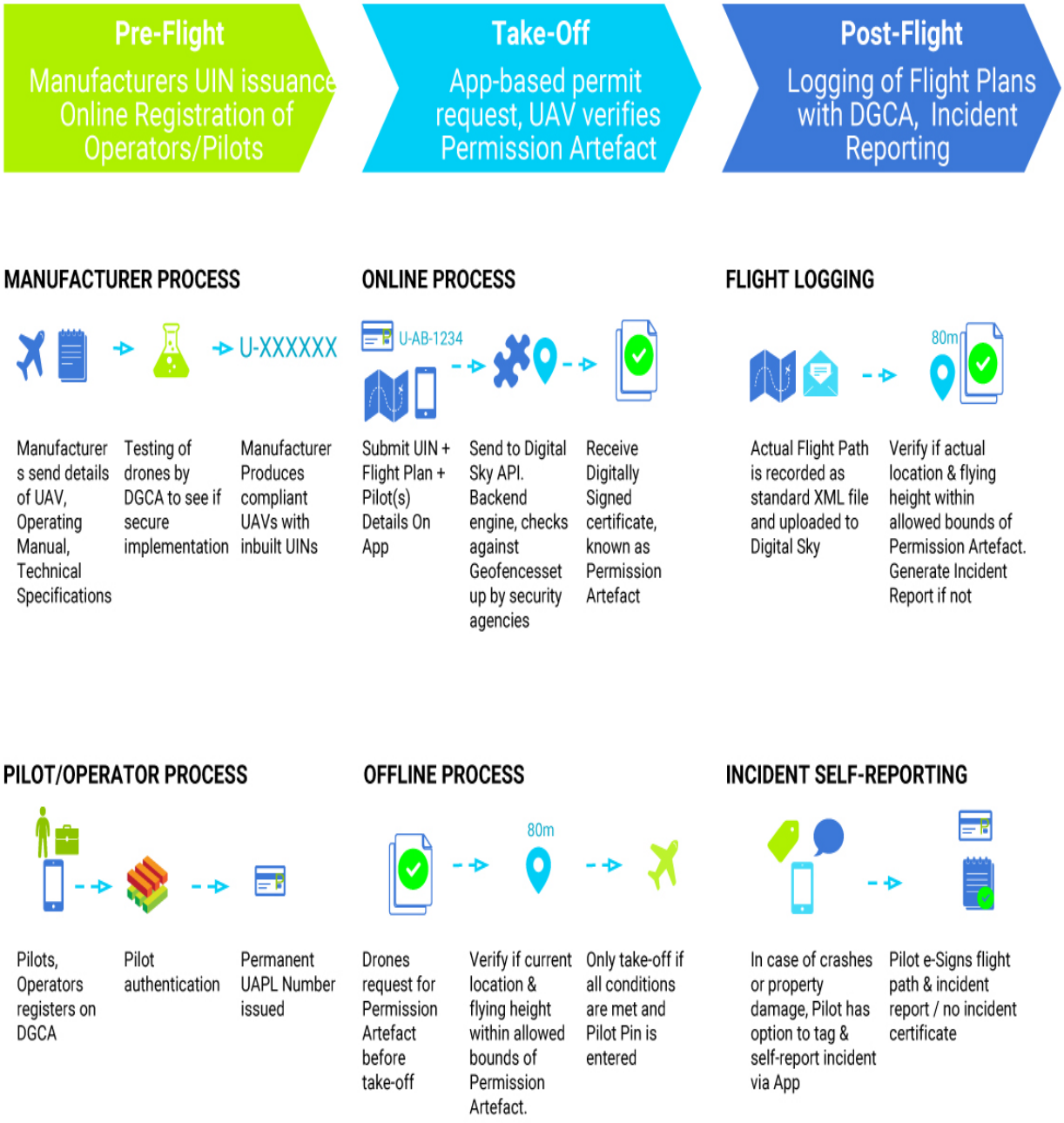
(b) **Operator Permit**. Operators need to obtain permits, ensuring they meet all regulatory requirements.

(c) **Instant Online Clearances**. Every flight needs an online clearance, which can be obtained instantly through the platform.

(d) **No Permission, No Take-off (NPNT)**. The platform enforces this policy, which means drones won’t take off without the required permission.

(e) **Geo-fencing and Tracking**. The system includes features for geo-fencing and tracking to ensure safe operations.

HOW DIGITAL SKY WORKS



Airspace Zones (Green, Yellow, And Red)

19. The airspace map of India shows red, yellow, and green zones across the country, allowing civilian drone operators to check the demarcated no-fly zones or where they need to undergo certain formalities before permitted to fly one.

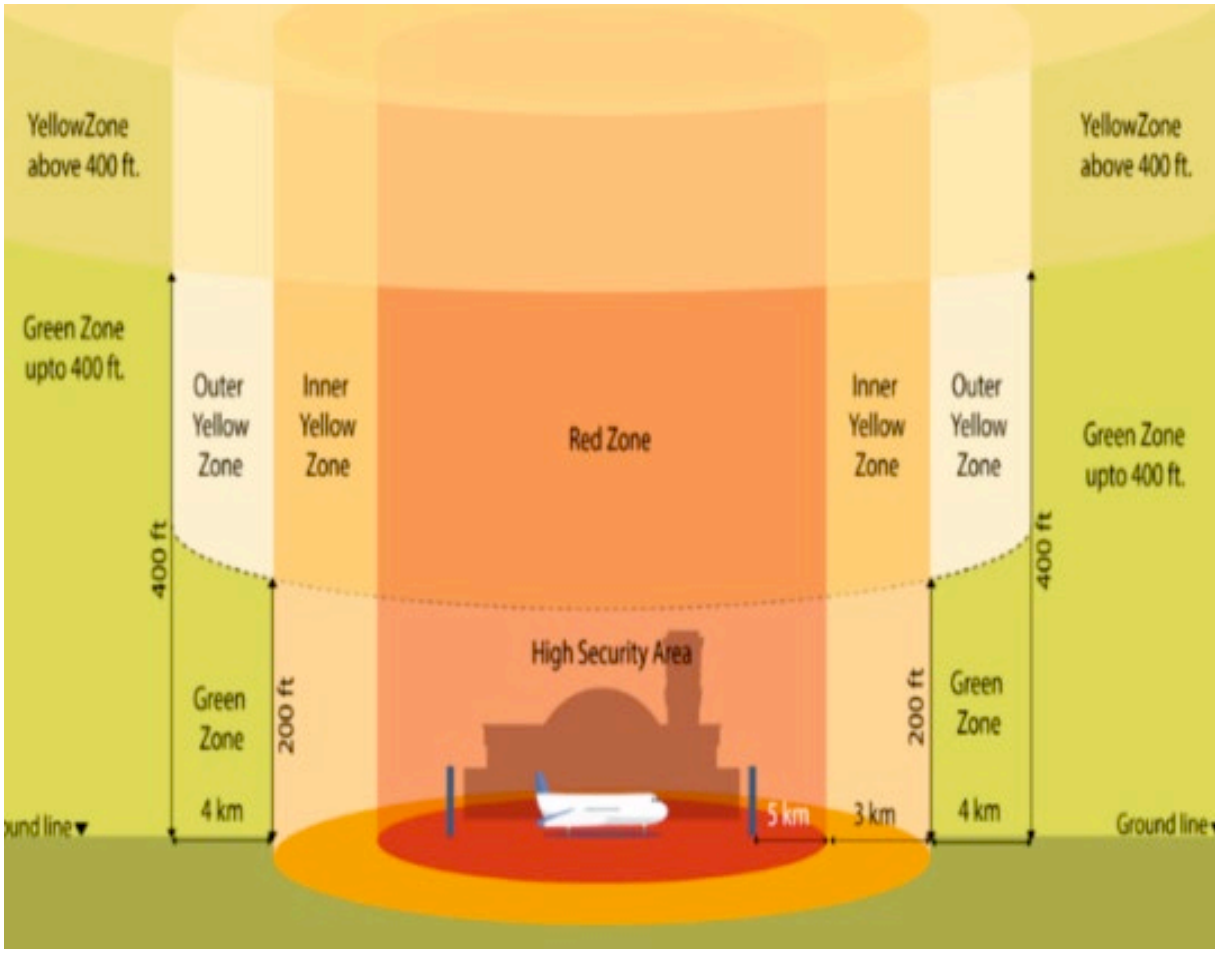
20. India has a well-defined airspace map for drone operations, which is available on the Digital Sky platform by the Directorate General of Civil Aviation. The airspace is divided into three zones:-

(a) **<u>Green Zone</u>**. Green Zone is an airspace up to 400 feet that is not designated as a red or yellow zone. No permission is required for operating drones in these areas.

(b) **<u>Yellow Zone</u>**. Airspace above 400 feet in a designated green zone, above 200 feet in areas located between 8-12 km from the perimeter of an operational airport, and above ground in areas located between 5-8 km from the perimeter of an operational airport is designated as 'Yellow Zone'. Permissions are required to operate in these zones.

(c) **<u>Red Zone</u>.** A complete no-fly zone where drone operations are prohibited without special authorisation is called a 'Red Zone'.

21. These zones help ensure safe and regulated drone operations across the country. If you are planning to fly a drone, make sure to check the latest airspace map on the Digital Sky platform.



<u>Insurance</u>

22. The provisions of the Motor Vehicles Act, 1988 (59 of 1988) and rules made thereunder shall apply to the third-party insurance of unmanned aircraft systems and compensation in case of damage to life or property caused by such an unmanned aircraft system, provided that a Nano unmanned aircraft system may operate without third-party insurance.

CONCLUSION

23. The registration and operation of unmanned aircraft systems (UAS) in India are governed by a comprehensive regulatory framework implemented by the Directorate General of Civil Aviation (DGCA).

KEY POINTS TO REMEMBER

- **Registration.** All UAS, except for Nano UAS, must be registered on the Digital Sky Platform to obtain a unique identification number.
- **Remote Pilot Certification.** Individuals operating UAS must possess a valid remote pilot license issued by the DGCA.
- **Airspace Zones.** India's airspace is divided into red, yellow, and green zones, with specific regulations governing drone operations in each zone.
- **Digital Sky Platform.** This online platform is crucial for registration, certification, flight planning, and other UAS-related activities.

24. By adhering to these regulations and utilising the Digital Sky platform, individuals can safely and legally operate UAS in India, contributing to the growth of this emerging technology.

SUMMARY

- **Mandatory Registration**. All UAS must be registered on the Digital Sky Platform and obtain a Unique Identification Number (UIN), unless specifically exempted.
- **Application Process**. Registration involves submitting Form D-2 on the Digital Sky Platform.
- **Transfer and De-Registration**. Transfer requires submission of Form D-3, and de-registration is necessary for a lost or permanently damaged UAS.
- **License Requirement**. Operating a UAS requires a valid remote pilot license, specifying the category and classification of the UAS.
- **Exemptions**. Nano UAS and micro UAS used for non-commercial purposes do not require a license.
- **Green Zone**. Up to 400 feet, no permission required.
- **Yellow Zone**. Requires permissions, especially near airports.
- **Red Zone**. It's a complete No-fly zone and Drones cannot fly without special authorisation.
- **Requirement**. Third-party insurance as per the Motor Vehicles Act, 1988, is mandatory for UAS, except for nano UAS operating without third-party insurance.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What is the primary platform for drone registration and management in India?

(a) DGCA Portal (b) Civil Aviation Website
(c) Digital Sky Platform (d) Drone Regulation Hub

Q2. Which form is required to apply for a unique identification number for an unmanned aircraft system?

(a) Form D-1 (b) Form D-2
(c) Form D-3 (d) Form D-5

Q3. What is required before transferring an unmanned aircraft system to another person?

(a) Unique Identification Number (UIN) and Fee (b) Written Permission
(c) Police Clearance (d) No formalities are needed.

Q4. A remote pilot license in India is valid for how many years?

(a) 5 years (b) 10 years
(c) 15 years (d) Lifetime

Q5. Which of the following is not required to hold a remote pilot license?

(a) Operating a nano unmanned aircraft system
(b) Operating a micro unmanned aircraft system for commercial purposes
(c) Operating a micro unmanned aircraft system for non-commercial purposes
(d) Operating a remote pilot organisation

Q6. At what age is an individual eligible to obtain a remote pilot license?

(a) 16 years (b) 18 years
(c) 21 years (d) 25 years

Q7. Which of the following does NOT apply to Nano unmanned aircraft systems?

(a) Remote pilot license requirement (b) Third-party insurance requirement
(c) Registration requirement (d) Both a and b

Q8. What is the maximum permissible altitude for drone operations in a green zone without permission?

(a) 100 feet (b) 200 feet

(c) 400 feet (d) 500 feet

Q9. Which of the following is true about red zones for drone operations?

(a) They are areas where drones can fly without any restrictions.

(b) No-fly zones unless special authorisation is granted.

(c) Only commercial drones are restricted.

(d) Drones can fly freely below 200 feet.

Q10. Which form is used to deregister a permanently lost or damaged drone?

(a) Form D-2 (b) Form D-3

(c) Form D-5 (d) Form D-7

Q11. Which feature ensures that drones won't take off without required permissions on the Digital Sky platform?

(a) Auto-pilot Mode (b) No Permission, No Take-off(NPNT)

(c) Geo-fencing (d) Flight Clearance

Q12. Which of the following is NOT a feature of the Digital Sky platform?

(a) Drone Registration (b) Geo-fencing

(c) Drone Insurance (d) Instant Online Clearances

Q13. To obtain a remote pilot license, what is the minimum educational qualification required?

(a) Class 8 (b) Class 10

(c) Class 12 (d) Graduate degree

Q14. Which airspace zone requires permission to operate above 400 feet?

(a) Green Zone (b) Red Zone

(c) Yellow Zone (d) White Zone

Q15. What is required to establish a Remote Pilot Training Organisation (RPTO)?

(a) Form D-2 (b) Form D-3

(c) Form D-4 (d) Form D-5

Q16. Which of the following organisations authorises remote pilot training organisations in India?

(a) Ministry of Civil Aviation

(b) Directorate General of Civil Aviation (DGC(A))

(c) Indian Air Force

(d) National Aviation Authority

Q17. What is the purpose of geo-fencing in drone operations?

(a) Prevent drones from entering restricted zones.

(b) Enable drones to fly faster.

(c) Improve drone camera quality

(d) Extend battery life

Q18. What is the required third-party insurance for drones based on?

(a) Drone size (b) Motor Vehicles Act, 1988

(c) Aviation law (d) Airspace zone

Q19. Which of the following drones are exempt from third-party insurance requirements?

(a) Nano unmanned aircraft systems

(b) Commercial drones

(c) Micro drones used for delivery

(d) All drones require third-party Insurance.

Q20. The validity period of authorisation to establish a remote pilot training organisation is:-

(a) 5 years (b) 7 years

(c) 10 years (d) Lifetime

True or False

Q1. Drones in the green zone can fly up to 400 feet without permission.

Q2. The Digital Sky platform is only used for drone registration and not for flight planning.

Q3. A remote pilot license is valid for a period of five years.

Q4. Operating a Nano unmanned aircraft system requires a remote pilot license.

Q5. No special authorisation is required to fly a drone in a red zone.

Q6. Geo-fencing helps prevent drones from entering restricted areas.

Q7. A drone transfer must be reported to the Digital Sky platform using Form D-3.

Q8. A person can establish a remote pilot training organisation without DGCA authorisation.

Q9. Micro unmanned aircraft systems used for non-commercial purposes require a remote pilot license.

Q10. Insurance for drones follows the Motor Vehicles Act, 1988.

Q11. In the yellow zone, permission is required to fly above 200 feet near an airport.

Q12. NPNT (No Permission, No Take-off) ensures drones won't take off without necessary permissions.

Q13. A drone license is required for all unmanned aircraft systems, including nano drones.

Q14. An individual must be at least 21 years old to apply for a remote pilot license.

Q15. De-registration of lost or damaged drones requires Form D-5 submission.

Short Answer Questions

Q1. What must all UAS be registered on?

Q2. What form is used to apply for UAS registration?

Q3. How can UAS be transferred?

Q4. What form is used for de-registration of a UAS?

Q5. What is required to operate a UAS?

Q5. What does the remote pilot license specify?

Q6. What are the eligibility criteria for obtaining a remote pilot license?

Q7. What form is used to apply for a remote pilot license?

Q8. How long is a remote pilot license valid?

Q9. Who can provide training for remote pilot licenses?

Q10. What form is used to apply for establishing an RPTO?

Q11. What platform manages UAS activities in India?

Q12. What are the three zones for drone operation in India?

Q13. What insurance requirements apply to UAS?

Long Answer Questions

Q1. Describe the general requirements and process for registering an unmanned aircraft system (UAS) on the Digital Sky Platform.

Q2. Explain the procedure and requirements for transferring and de-registering a UAS.

Q3. Discuss the remote pilot certification process, including eligibility, application, validity, and exemptions.

Q4. Analyse the role and importance of the Remote Pilot Training Organisation (RPTO) in the certification process.

Q5. Provide an overview of the Digital Sky Platform, including its functions in drone registration, operator permits, online clearances, and geo-fencing.

Q6. Describe the three zones for drone operation in India and the specific permissions required for each.

Q7. Explain the insurance requirements for UAS in India and any exemptions that apply.

DRONES

CHAPTER D IV : APPLICATION OF DRONES



TEACHING INSTRUCTIONS

Period : One (01)

Type : Lecture and Presentation

Year : 3rd Year SD/SW

Conducting Officer : Officer of the Unit

Training Aids : Classroom, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, and Models.

Time Plan

- Introduction : 05 min
- Application of Drones in Various Fields : 10 min
- Search & Rescue and Disaster Response : 10 min
- Career Opportunities in Drone Industry : 08 min
- Q & A Session and Conclusion : 07 min

INTRODUCTION

1. Drone is a very versatile equipment because they can be flown remotely. It has an exceptional quality of taking aerial photos and videos, giving a splendid view from the sky. Plus, they are handy for delivering packages, especially to those in hard-to-reach places. These are just a few examples of the many uses of drones that make them immensely valuable.

2. Farmers now use drones to check their crops from a bird's eye view. These are the changing technology. And it is not just for them-drones are also used in search and rescue mission to find people in difficult spots. This new use of drones is changing farming and emergency response incredibly.

PREVIEW

The lecture will be conducted in the following parts:-

- Part I : Application of drones in various fields.
- Part II : Search & Rescue and Disaster Response.
- Part III : Career Opportunities in
- Drone Industry

LEARNING OBJECTIVES

- To know the application & usage of drones.
- To know how the drones are helpful in search & rescue.

PART I : APPLICATION OF DRONES

3. Drones can reach hard-to-access or dangerous areas, inspect wind turbines, monitor construction sites, and survey agricultural fields. They often have cameras, GPS, and sensors to capture high-quality images and data. This makes them useful for precision tasks in fields like real estate, media, public safety, and many more things.

4. Use of drones is on the rise, with industries finding innovative ways to utilise this technology. Other than military, some of the most common uses of drones are given in succeeding paragraphs.

Agriculture

5. **Crop Monitoring and Health Assessment**. Due to the efficacy of drone applications, farmers can now easily check their crops' health and look out for pests and diseases. These gadgets have special cameras that give detailed information about plant health. This technology helps farmers quickly spot problem areas and act fast.

6. **Precision Farming**. Drones with sensors and cameras can take detailed, high-resolution pictures of crops. These photos help farmers monitor crop health, spot pests, and catch diseases early. By reviewing this data, farmers can decide where to spray pesticides, use fertilisers, or water their crops. This boosts productivity, reduces waste, and lessens environmental impact.



Environmental Monitoring and Conservation

7. **Wildlife Tracking and Poaching Prevention**. Drones are making a difference in monitoring wildlife and stop poaching. They can cover huge areas and gather real-time data without bothering the animals, which helps conservationists monitor wildlife's movements and behaviour. Drones with thermal cameras can spot poachers, especially at night, allowing authorities to act quickly against illegal activities. This technology is a revolutionary innovation for protecting endangered species because it makes the surveillance extra efficient and less intrusive.

8. **Forest Management**. When it comes to forest management, drones are transformative innovations. They can fly over vast areas to check on tree health, watch for deforestation, and spot illegal logging. With Light Detection and Ranging (LiDAR) technology, drones create detailed 3D maps of forests, showing the canopy structure and forest biomass. This information is key for sustainable forest management, making planning and conservation much easier. Drones can also speed up reforestation by spreading seeds over large, hard-to-reach areas. It is a faster, more efficient way to plant trees.

Infrastructure Inspection and Construction

9. **Building and Structural Inspections**. Drones are changing the perspective of inspecting the buildings and structures. They are making it faster, safer, and cheaper. Instead of using scaffolding, ladders, or even helicopters, drones can easily reach those hard-to-get spots and capture clear images and videos. This helps engineers and inspectors find issues like cracks or rust without being there in person. Plus, it keeps the inspection thorough and safe, reducing risks and keeping things smooth.

10. **Construction Site Monitoring**. Keeping track of construction progress is important for meeting deadlines and ensuring quality work. Drones can help a lot with this. They provide a bird's-eye view of the entire site, letting project managers see a real time image. By doing regular aerial surveys, they can capture photos and videos that show how the project is going and highlight any issues. This technology also helps teams communicate better since everyone gets clear visual data. Besides, it helps manage inventory by checking materials on-site, ensuring resources are used smartly, and timelines are met.



Energy Sector

11. **Wind Turbine Inspection**. Wind turbines need regular check-ups to ensure they are working safely. Drone use cases for these inspections are remarkable. Drones can fly close to the turbines, taking high-resolution photos and videos of the blades and other parts. This helps engineers quickly spot any damage or wear and tear without having to climb up those tall structures. The whole process becomes faster, safer, and more accurate. Repairs can be done quickly, thereby reducing downtime and keeping the turbines running smoothly for continuous energy.

12. **Solar Panel Inspections.** It is important to maintain solar panels properly to keep them working well. Drones with thermal cameras can help by quickly scanning large solar farms and finding issues like hotspots from damage or debris. Catching these problems early, prevents energy loss and keeps everything running efficiently. This technology saves time and reduces the need for manual checks, allowing technicians to fix issues faster. Overall, it helps solar farms produce clean energy reliably.

Media and Entertainment

13. **Event Coverage**. Drones are transforming how we cover events in media and entertainment. They give us unique aerial views that used to be expensive or impossible. At concerts, sports, or festivals, they capture wide shots of the crowd and stage, making viewers feel more involved. They can quickly move around to cover different parts of an event, providing real-time footage for live broadcasts. This flexibility helps media companies create more engaging content.

14. **Cinematography**. In filmmaking, drones have become a partner for cinematographers. They let you capture stunning aerial shots without needing cranes or helicopters, which cuts costs and sparks creativity. You can fly them through narrow spaces, hover at exact angles, and get smooth tracking shots, all of which boost visual storytelling. This means even small teams can now achieve amazing scenes that were tough with traditional setups. Overall, drones have made aerial cinematography more accessible and have added a tonne of flexibility and creativity to the industry.



Real Estate and Property Management

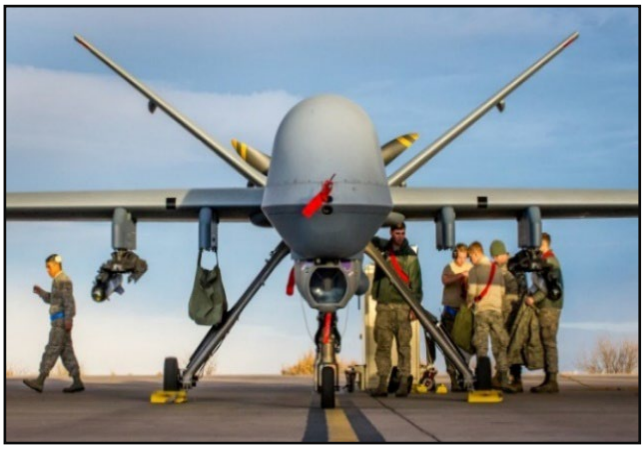
15. **Property Showcases**. Drones have changed real estate. They give an incredible aerial view, showing buyers the whole property and its surroundings. This helps buyers see the neighbourhood, yard, and other details that regular photos might miss. These shots make listings more appealing and help properties sell faster. Agents can use them to show features like pools, gardens, and nearby amenities, giving a better overall view of the property.

16. **Land Survey**. Accurate land surveying is key in real estate. Drones make this process a lot easier by quickly capturing detailed data. They can map large areas faster than traditional methods, helping developers make quicker decisions and save time and money. They handle tough terrain well, so no part of the land gets missed. Using this technology gives precise and reliable data, which is essential for planning and development.

Drones in the Military

17. Drones were created mainly for military purposes. Their use in all other areas that we have discussed above is but secondary. Countries at war have been using drones for a substantial period of time, dating back as early as World War I. Use of drones made a lot of sense in wars given the fact that the lives of pilots don't have to be risked.

18. Today, drones are available in many forms and sizes and are used by numerous countries around the world. They have played a role in predicting conflicts and also in helping countries fight wars with fewer casualties.





Commercial Uses of Drones

19. Commercial use of drones is on the rise, with industries finding innovative ways to utilise this technology. Some of the most common commercial uses of drones include:-

(a) **Aerial Photography and Videography**. Drones have revolutionised film and TV production.

(b) **Construction and Infrastructure Inspection**. In construction, drones make surveying land, monitoring progress, and inspecting structures easy and fast.

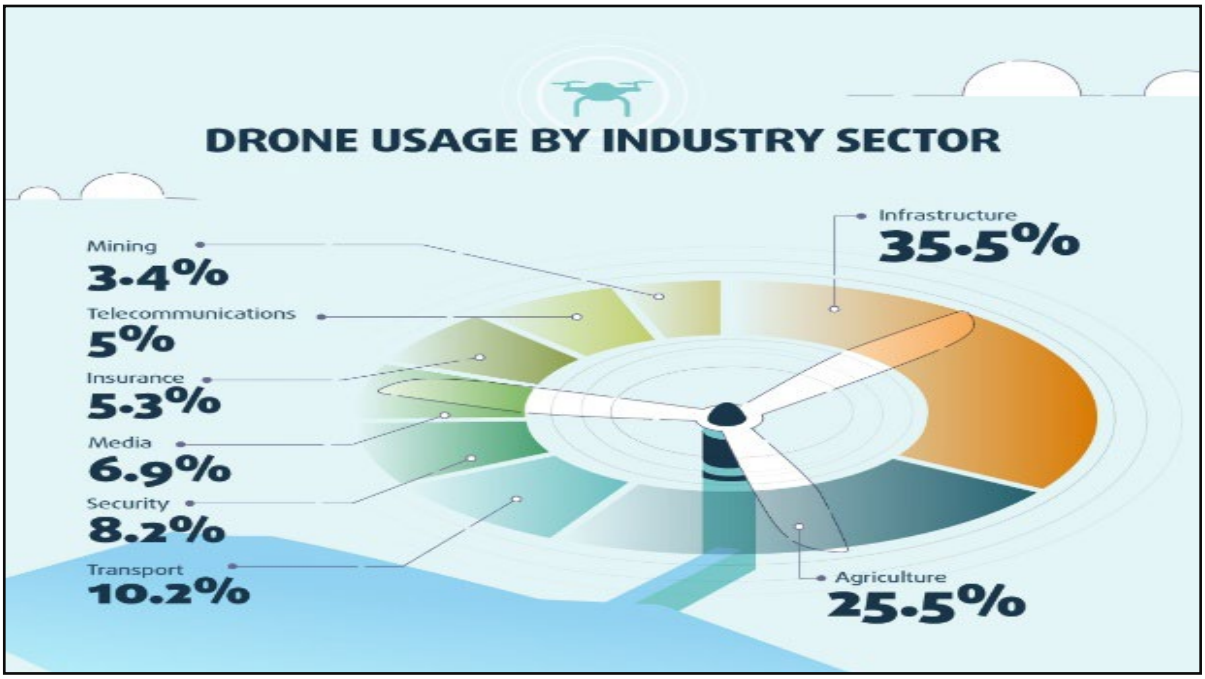
(c) **Agriculture**. Drones are revolutionising agriculture by improving how we manage crops.

(d) **Energy Sector**. Drones are changing the energy sector by making maintenance and inspections easier and cheaper.

(e) **Environmental Monitoring and Conservation**. Drones are an asset for environmental monitoring and conservation.

(f) **Delivery and Logistics**. Companies like Amazon are testing drone delivery services and integrating these flying robots into delivery and logistics.

(g) **Real Estate and Property Management**. Drones can provide detailed aerial footage of properties, giving potential buyers or renters a better understanding of the space.



Mining and Surveying. The mining industry has been revolutionised by drones, offering a safer and more efficient method to survey and map mining sites.

Key Benefits of Using Drones

20. **Cost-effective**. Drones are much more affordable than traditional methods like helicopters or cranes, making them accessible to smaller production teams and industries.

21. **Efficiency**. Drones can cover large areas quickly and provide high-resolution images and data in a fraction of the time as compared to traditional methods. This saves time, resources, and labour costs.

22. **Versatile**. Drones can be equipped with different cameras and tools depending on the specific needs of each industry. This versatility makes them useful in various applications such as aerial photography, mapping, surveying, delivery, etc.

23. **Safe**. Drones eliminate the need for workers to access dangerous or hard-to-reach areas, making operations safer for everyone involved. They also are equipped with advanced collision avoidance technology to prevent accidents.

24. **Eco-friendly.** Drones reduce the need for traditional methods that emit harmful pollutants, making them a more environmentally friendly option for various industries.

PART II : DISASTER MANAGEMENT AND RESPONSE

Search and Rescue Operations

25. Drones are super advanced now, with high-resolution cameras and thermal imaging. They can quickly scan big areas to find survivors, which is great when traditional methods are too slow or risky. For example, after an earthquake, they can help rescuers find trapped people by detecting body heat. Plus, they can access hard-to-reach or dangerous places, letting rescue teams provide help fast without risking more lives.

Damage Assessment

26. After a disaster, it is really important to quickly check the damage to plan recovery. Drones can fly around and take detailed pictures, giving authorities a fast overview. This helps understand the impact and prioritise efforts. Using drones is quicker and more accurate than doing inspections on foot, leading to faster decisions and quicker resource deployment where needed most. Drones offer great support for efficient disaster response.

Public Safety and Law Enforcement

27. **Crime Scene Analysis**. Drones are changing how we analyse crime scenes by providing detailed aerial views. They take high-resolution images from various angles, helping us spot details we might miss on the ground. This helps preserve evidence and document it efficiently, assisting law enforcement in building stronger cases and solving crimes faster. Besides, use of Drones mean we disturb the crime scene less, keeping its integrity intact.

28. **Crowd Monitoring**. Handling large crowds at events can be tough for police, but drones make it easier. They fly overhead, watch crowd movements in real-time, and give a view that ground officers can’t get. This helps spot issues like overcrowding or disturbances quickly, so responses are faster. Drones improve public safety and help

events run smoothly. They are cost-effective and reduce the need for lots of police, making the atmosphere more comfortable for everyone.

29. **Public Safety**. Public safety agencies like police, fire departments, and search and rescue teams find drones very useful. With thermal cameras, they can locate missing persons or detect fire hotspots. This helps emergency personnel respond quickly and safely. In disasters, drones give real-time aerial views to assess damage, find survivors, and map affected areas. They can also deliver medical supplies or rescue gear to hard-to-reach places. This is a critical drone use that enhances public safety operations.

Unmanned Combat Aerial Vehicle (UCAV)

30. An unmanned combat aerial vehicle (UCAV), also known as a combat drone, fighter drone, or battlefield UAV, is an unmanned aerial vehicle (UAV) that is used for intelligence, surveillance, target acquisition, and reconnaissance. It can also carry carries aircraft ordnance such as missiles, anti-tank guided missiles (ATGMs), and/or bombs for the strike.

31. **Weapon Systems Used in Drones**. Drones, especially those used in military operations, can be equipped with a variety of weapon systems. Some common types are as follows:-

(a) **Missiles**. Drones can carry air-to-surface missiles, such as the Hellfire missile, which is designed for precision strikes against ground targets.

(b) **Bombs**. Various types of bombs, including guided bombs and cluster bombs, can be deployed from drones.

(c) **Anti-Tank Missiles**. Drones can be equipped with Anti-Tank Guided Missiles (ATGM) to target armoured vehicles.

(d) **Autocannons and Machine Guns**. Some drones are armed with autocannons and machine guns for close-range engagements.

(e) **Laser Guided Weapons**. These weapons use laser targeting to hit specific targets with pinpoint accuracy.

(f) **Incendiary Devices**. Drones can carry incendiary devices designed to start fires or destroy specific targets.

32. These weapon systems enable drones to carry out a range of missions from precision strikes to close air support.

Anti-Drone Systems

33. Anti-Drone Systems, also known as Counter-Unmanned Aerial Systems(C-UAS), are designed to detect, track, and neutralise unauthorised malicious drones. Some common anti-drone systems are:-

(a) **RF Jamming**. RF refers to Radio Frequency. Devices like Drone Gun Tactical and Drone Defender V2 emit radio frequency signals to disrupt the communication links between the drone and its operator, forcing the drone to land or return to its home point.

(b) **High Energy Laser Weapon System (HELWS)**. Systems like Raytheon's High Energy Laser Weapon System use laser beams to disable or destroy drones by targeting their sensitive components.

(c) **Interceptor Drones**. Autonomous drones equipped with nets and other capture mechanisms that can physically engage and disable rogue drones in mid-air.

(d) **GPS Spoofing**. Systems that manipulate GPS signals to mislead the drones' navigation system causing it to lose control.

(e) **Projectiles**. Some systems use projectiles such as nets or missiles to physically capture or destroy drones.

34. These technologies are crucial for maintaining security and safety in various environments, from airports to critical infrastructure.

PART III : CAREER OPPORTUNITIES IN DRONE INDUSTRY



35. The drone industry is rapidly expanding, opening up a wide range of exciting career opportunities across various sectors. Here are some of the key areas where drone technology is making a significant impact and creating job opportunities.

36. **Drone Pilot**.

(a) **Role**. Operate drones for photography, videography, surveying, mapping, inspections, deliveries, and more.

(b) **Industries**. Real estate, construction, agriculture, film and media, public safety, and logistics.

37. **Drone Engineer**.

(a) **Role**. Design, develop, and maintain drones and their components.

(b) **Industries**. Aerospace, robotics, and drone manufacturing companies.

38. **Drone Data Analyst**.

(a) **Role**. Analyse and interpret data collected by drones, such as aerial imagery and LiDAR data.

(b) **Industries**. Agriculture, urban planning, environmental conservation, and insurance.

39. **Drone Software Developer**.

(a) **Role**. Develop software for drone control, automation, and data processing.

(b) **Industries**. Drone manufacturing companies, software development firms, and research institutions.

40. **Drone Instructor**.

(a) **Role**. Train individuals on drone piloting, maintenance, and regulations.

(b) **Industries**. Aviation schools, drone training academies, and corporate training programs.

41. **Drone Consultant**.

(a) **Role**. Provide expert advice on drone technology, regulations, and business applications.

(b) **Industries**. Various industries seeking to adopt drone technology.

42. **Drone Technician**.

(a) **Role**. Perform maintenance, repairs, and troubleshooting on drones.

(b) **Industries**. Drone service centres, rental companies, and drone manufacturers.

43. **Drone Traffic Manager**.

(a) **Role**. Manage and coordinate drone traffic in urban and rural areas.

(b) **Industries**. Aviation authorities and drone service providers.

44. **Drone Delivery Specialist**.

(a) **Role**. Deliver goods and packages using drones.

(b) **Industries**. E-commerce, logistics, and healthcare.

45. **Drone Filmmaker/Photographer**.

(a) **Role**. Capture stunning aerial footage and images using drones.

(b) **Industries**. Film and television production, advertising, and real estate.

46. **Additional Career Opportunities.**

(a) **Drone Law Enforcement.** Assisting in law enforcement operations, search and rescue missions, and border patrol.

(b) **Drone Agriculture**. Monitoring crops, spraying pesticides, and delivering seeds.

(c) **Drone Inspection**. Inspecting infrastructure, power lines, and oil pipelines.

(d) **Drone Surveying and Mapping**. Creating detailed maps and 3D models of land and buildings.

47. To pursue a career in the drone industry, consider obtaining a drone pilot license, relevant certifications, and practical experience. Stay updated on the latest drone technology and industry trends to enhance your career prospects.

CONCLUSION

48. Drones have emerged as a versatile tool with applications across various industries and sectors. From agriculture and construction to media and entertainment, drones are revolutionising the way we work and live.

49. **Key Benefits of Drone Technology**:-

(a) **Efficiency**. Drones can cover vast areas quickly and efficiently, saving time and resources.

(b) **Safety**. Drones can be used to inspect dangerous or inaccessible areas, reducing the risk to human life.

(c) **Data Collection**. Drones can collect high-resolution data, such as aerial photography and thermal imaging, to inform decision-making.

(d) **Precision**. Drones can perform tasks with precision, such as spraying on crops or delivering packages.

HIGHER-ORDER THINKING SKILLS (HOTS)

- **Innovative Solutions**. Consider yourself in charge of creating a drone-based solution for managing traffic. How would you design the system to maximise traffic flow and reduce congestion? What factors would you consider, and why?

- **Ethical Implications**. Discuss the ethical considerations of using drones for public surveillance. How can privacy concerns be balanced with the benefits of enhanced security and crime prevention?

- **Environmental Impact**. Analyse the potential environmental impact of large-scale drone delivery services. What measures could be implemented to mitigate any negative effects, and how would these measures align with sustainable practices?

- **Interdisciplinary Integration**. How can drones be integrated into existing emergency response systems to improve efficiency and effectiveness? Propose a comprehensive plan that includes technological, logistical, and regulatory aspects.

- **Technological Advancements**. Evaluate the current limitations of drone technology in commercial applications, such as delivery services and agriculture. What technological advancements would you prioritise to overcome these limitations, and why?

- **Societal Impact**. Assess the societal implications of widespread drone usage in rural and remote areas. How can drones be leveraged to address challenges such as healthcare access, education, and infrastructure development?

- **Regulatory Frameworks**. Propose a regulatory framework for the use of drones in airspace shared with manned aircraft. How would you ensure safety and efficiency while promoting innovation and growth in the drone industry?

- **Global Applications.** Explore the potential applications of drones in developing countries. How can drones contribute to economic development, infrastructure improvement, and disaster management in these regions?

- **Future Prospects**. Envision the future of drone technology over the next decade. What new applications do you foresee emerging, and how will these advancements transform industries and daily life?

SUMMARY

- **Agriculture**. Farmers use drones for crop monitoring, spraying pesticides, and analysing the field conditions. It is all about improving efficiency and yields.

- **Delivery Services**. Companies like Amazon are experimenting with drones for fast and efficient package delivery. Imagine getting your order in minutes.

- **Environmental Monitoring**. Drones help monitor wildlife, track deforestation, and survey areas affected by natural disasters. They provide crucial data without disturbing the ecosystems.

- **Entertainment**. In filmmaking, drones enable breath taking aerial shots that were once only possible with helicopters. They are also used in sports broadcasting and creating light shows.

- **Military**. Drones are used for reconnaissance, surveillance, and targeted strikes. They have transformed modern warfare but at the same time also raised ethical and legal questions.

- **Healthcare**. In remote areas, drones can deliver medical supplies, vaccines, and even blood. They offer a lifeline where traditional transport cannot reach.

- **Infrastructure Inspection**. Drones are used to inspect bridges, power lines, and pipelines. They can access hard-to-reach places and provide detailed images, ensuring timely maintenance and safety.

- **Personal Use**. Many people use drones for photography, racing, and also as a hobby. It is a fun way to explore and capture the world from new angles.

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. Which industry uses drones to monitor crop health and boost yields?

(a) Healthcare (b) Agriculture
(c) Entertainment (d) Military

Q2. Which company is known for experimenting with drone delivery services?

(a) Google (b) Microsoft
(c) Amazon (d) Apple

Q3. In environmental monitoring, what is one key use of drones?

(a) Delivering packages (b) Filming movies
(c) Tracking wildlife and deforestation (d) Racing

Q4. What kind of drones does the military typically use reconnaissance for and surveillance?

(a) Delivery drones (b) Racing drones
(c) Surveillance drones (d) Filmmaking drones

Q5. Which industry uses drones for inspecting bridges, power lines, and pipelines?

(a) Agriculture (b) Healthcare
(c) Infrastructure inspection (d) Retail

Q6. Drones are used in filmmaking primarily for what purpose?

(a) Delivering scripts (b) Providing aerial shots
(c) Monitoring audience reactions (d) Transporting equipment

Q7. What kind of drones are used for entertainment purposes, like light shows and racing?

(a) Delivery drones (b) Agricultural drones
(c) Hobby drones (d) Military drones

Q8. In the healthcare industry, how are drones being utilised?

(a) Delivering medical supplies to remote areas
(b) Performing surgeries
(c) Monitoring patient health
(d) Administering vaccines directly

Q9. Which application involves drones performing tasks such as spraying pesticides and mapping fields?

(a) Retail
(b) Agriculture
(c) Military
(d) Entertainment

Q10. How are drones contributing to disaster response efforts?

(a) Delivering food
(b) Surveying damaged areas and aiding in search and rescue operations
(c) Recording news footage
(d) Providing internet access

Q11. Which type of drones are used to capture stunning aerial footage for movies?

(a) Delivery drones
(b) Filmmaking drones
(c) Military drones
(d) Agricultural drones

Q12. In logistics, what is a primary use of drones?

(a) Transporting employees
(b) Delivering packages quickly and efficiently
(c) Monitoring warehouses
(d) Managing inventory

Q13. Which sector uses drones to provide inspection of infrastructure like power lines and towers?

(a) Agriculture
(b) Entertainment
(c) Infrastructure inspection
(d) Military

Q14. What is a common use of drones in the environmental sector?

(a) Planting trees
(b) Monitoring climate change effects
(c) Building habitats
(d) Introducing new species

Q15. In what context are drones used to provide realtime surveillance and intelligence?

(a) Agriculture
(b) Healthcare
(c) Military operations
(d) Entertainment

Fill in the Blanks

Q1. Drones are used in_____ to monitor crop health and increase yields.

Q2. Companies like_____ are experimenting with drone delivery services.

Q3. In environmental monitoring, drones are used to track _____ and deforestation.

Q4. The _____industry uses drones for providing stunning aerial shots movies.

Q5. Drones used for surveying damaged areas and aiding in _____ operations are crucial in disaster response.

Q6. In the healthcare industry, drones deliver _____ supplies to remote areas.

Q7. For infrastructure inspection, drones provide detailed images of _____, power lines and pipelines.

Q8. Drones used for personal _____ include photography and racing drones.

Q9. In logistics, drones are used to deliver _____ quickly and efficiently.

Q10. The _____ sector uses drones for realtime surveillance and intelligence.

Short Answer Questions

Q1. How are drones used in agriculture?

Q2. Which company is experimenting with drones for fast and efficient package delivery?

Q3. What environmental applications do drones have?

Q4. How are drones utilised in filmmaking?

Q5. Name two military uses of drones.

Q6. What healthcare services can drones provide in remote areas?

Q7. How do drones assist in infrastructure inspection?

Q8. What are some personal uses of drones?

Long Answer Questions

Q1. Discuss the various ways drones are used in agriculture to improve efficiency and yields.

Q2. Explain how companies like Amazon are using drones for delivery services and the potential benefits.

Q3. Describe the role of drones in environmental monitoring, including their advantages.

Q4. Analyse how drones have revolutionised the entertainment industry, particularly in filmmaking and sports broadcasting.

Q5. Evaluate the impact of drones on modern warfare, considering both their benefits and the ethical/legal questions they raise.

Q6. How do drones contribute to healthcare delivery in remote areas, and what are the potential impacts?

Q7. Outline the advantages of using drones for infrastructure inspection and maintenance.

Q8. Explore the various personal uses of drones and how they have changed the way people interact with technology for hobbies and photography.

DRONES

CHAPTER D V : DRONE OPERATIONS



TEACHING INSTRUCTIONS

Periods : Two (02)

Type : Lecture and Presentation

Year : 3rd Year SD/SW

Conducting Officer : Officer/Trained PI/ AMI/ CGI

Training Aids : Classroom, Computer with OHP, Screen, Pointer Staff, Presentation, Script or Book Flagged or Lesson Plan in File, Board and Markers, and Models.

Time Plan

- Introduction : 05 min
- Pre-Flight Checks : 35 min
- Start Up & Controls : 20 min
- Emergencies : 12 min
- Q & A Session and Conclusion : 08 min

INTRODUCTION

1. Drones, also known as unmanned aerial vehicles (UAVs), have revolutionized various industries with their versatility and precision. From aerial photography and surveying to military operations and package delivery, drones are being utilized in a multitude of applications. Understanding the operations of drones is crucial to harnessing their potential and ensuring safe and efficient use. This chapter will delve into the basics of drone operations, covering aspects such as pre-flight checks and preparations, controls, manoeuvre and emergency procedures.

PREVIEW

The lecture will be conducted in the following parts:-

- Part I : Pre-Flight Checks
- Part II : Start Up & Controls
- Part III : Emergencies

LEARNING OBJECTIVES

- To know the pre-flight checks
- To learn Starting Up, Controls, and how drone is manoeuvred through the air
- To know and understand the emergencies relating to drones

PART I : PRE-FLIGHT CHECKS

2. Pre-flight activities are the duty of the Remote Pilot in Command (RPIC) before the start of the flight operation. Activities include inspection of the aircraft, assessment of the operating location, briefing crew members involved in the operation, and equipment checkouts.

Drone Pre-Flight Checklist

3. **Weather and Site Safety Check**.

(a) Chance of precipitation less than 10%

(b) Wind speed under 15 knots (less than 20 mph or 32 km per hour)

(c) Cloud base at least 500 feet

(d) Visibility at least 5 km

(e) If flying at dawn/dusk, double-check civil twilight hours.

(f) Establish take-off, landing, and emergency hover zones.

(g) Potential for electromagnetic interference?

(h) Look for towers, wires, buildings, trees, or other obstructions.

(j) Look for pedestrians and/or animals and set up a safety perimeter if needed.

(k) Discuss the flight mission with other crew members if present.

4. **Airspace Research**.

(a) Conduct Airspace Classification Review.

(b) Check for NOTAMs (Notices to Airmen) and TFRs (Temporary Flight Restrictions).

(c) Secure Airspace Authorisation (if required).

(d) If airspace authorisation is required, make sure you have approval.

5. **Batteries, Updates, and Connectivity**.

(a) Make sure drone batteries and remote-control batteries are fully charged.

(b) Pack extra batteries as needed.

(c) Make sure your drone and remote control have the latest firmware update installed.

(d) Check your SD card(s) to make sure you have enough memory for the mission (if relevant).

(e) Make sure you have a USB cable for your mobile phone or tablet.

(f) Make sure you will have connectivity in the field.

6. **Visual Aircraft & System Inspection**.

(a) The registration number is displayed properly and is legible.

(b) Look for abnormalities-aircraft frame, propellers, motors, undercarriage.

(c) Look for abnormalities-gimbal, camera, transmitter, payloads, etc.

(d) Gimbal clamp and lens caps are removed.

(e) Clean lens with a micro-fibre cloth.

(f) Attach propellers, battery/fuel source, and insert SD card/lens filters.

7. **Powering Up**.

(a) Turn on the remote control and open up the DJI Go 4 application.

(b) Turn on aircraft

(c) Verify established connection between transmitter and aircraft

(d) Position antennas on the transmitter towards the sky.

(e) Verify the display panel/FPV screen is functioning properly.

(f) Calibrate Inertial Measurement Unit (IMU) as needed.

(g) Calibrate the compass before every flight.

(h) Verify battery/fuel levels on both transmitter and aircraft.

(j) Verify that the UAS has acquired GPS location from at least six satellites.

8. **Taking Off**.

(a) Take off to eye-level altitude for about 10-15 seconds.

(b) Look for any imbalances or irregularities.

(c) Listen for abnormal sounds.

(d) Pitch, roll, and yaw to test control response and sensitivity

(e) Check for electromagnetic interference or other software warnings.

(f) Do one final check to secure the safety of the flight operations area.

(g) Proceed with flight mission.

PART II : START UP AND CONTROLS

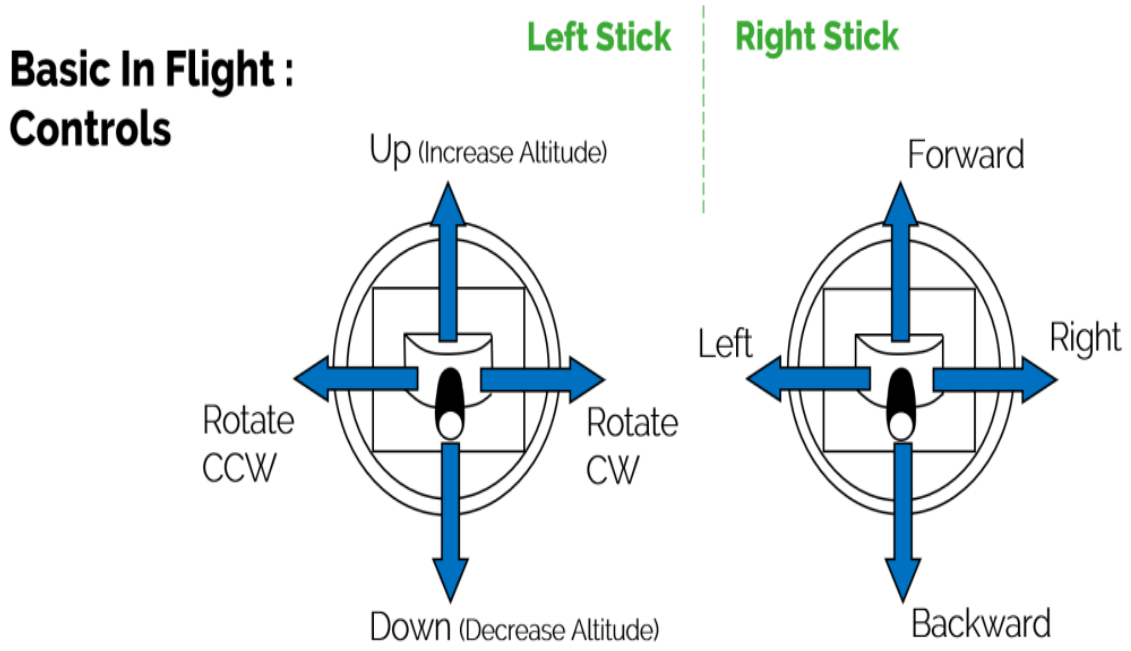
9. The four main drone controls are achieved by:-

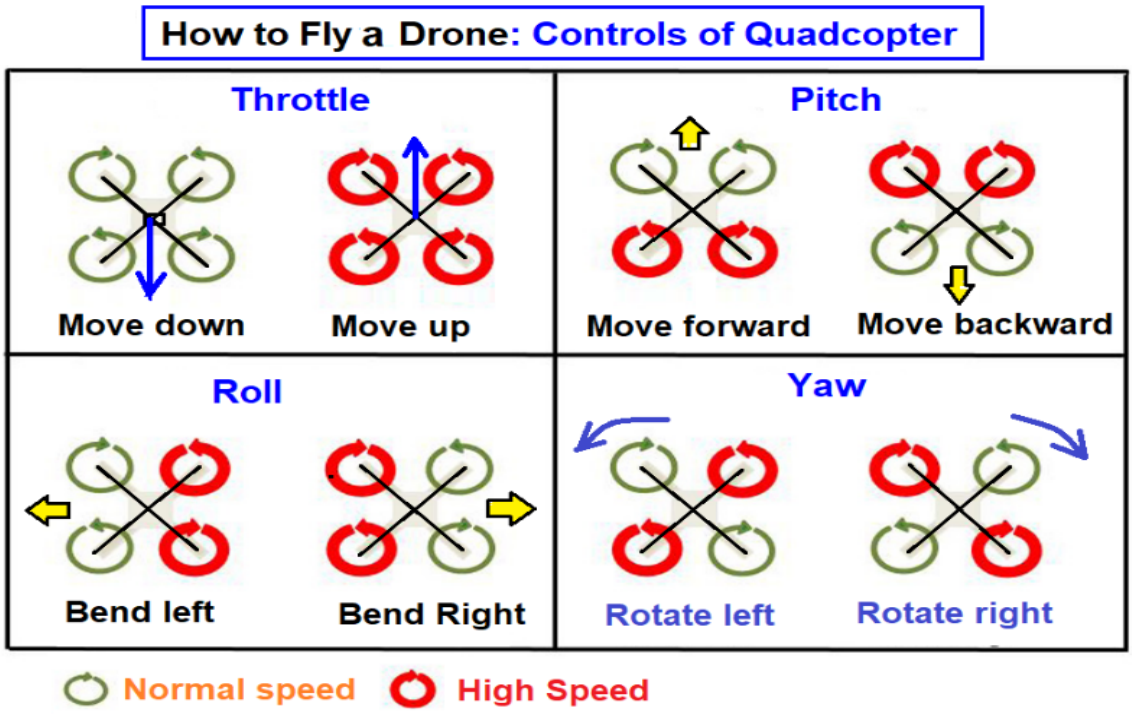
(a) **Roll**. Done by pushing the right stick to the left or right. Literally rolls the drone, which manoeuvres the drone left or right.

(b) **Pitch**. Done by pushing the right stick forwards or backward. Tilts the drone, which manoeuvres the drone forwards or backward.

(c) **Yaw**. Done by pushing the left stick to the left or to the right. Rotates the drone left or right. Points the front of the copter in different directions and helps with changing directions while flying.

(d) **Throttle**. To increase, push the left stick forwards. To decrease, pull the left stick backward. This adjusts the altitude, or height, of the drone.





PART III : EMERGENCY PROCEDURES

10. Ensuring the safety of RPAS operations requires an understanding of associated risks or emergencies and their respective effects. The table below highlights potential RPAS-related emergency operations, its plausible effects, and appropriate contingency/ failsafe measures we have put in place to safeguard the same.

SER NO	EMERGENCY/ RISK/POTENTIAL HAZARD	POSSIBLE EFFECTS	CONTINGENCY PLAN/ FAILSAFE MEASURE
(a)	**Aircraft Fly-away/ Geofence non-conformance**	• Mid-air collision with another manned or unmanned aerial aircraft. • Crash into buildings/ obstacles. • Crash debris injuring people on the ground.	Geo-fencing is always enabled and doesn't allow the pilot to fly outside the defined corridor.
(b)	**Battery Power Loss or Failure**	• Uncontrolled descent or landing could cause serious injury to many persons on the ground and possible fatalities. • Chances of mid-air fire leading to damage of ground assets. • Crash landing.	Battery failsafe cautions the pilot of a low battery warning and immediately initiates a return to home (RH) and lands the aircraft autonomously at a predefined location.
(c)	**Loss of Communication/ Control link**	• RPAS crash into one or more buildings. • Obstacles resulting in widespread secondary injury to humans from UAS debris and/or building damage. • Uncontrolled descent.	Communication failure will also trigger the smart RH, where the aircraft returns at a safe altitude.
(d)	**Unsuccessful Landing/ Flight Termination**	• Vehicle break-up or totalling. • Post-crash fire that injures ground crew/threatens wildlife and environment. • RPAS collides with buildings/bridges/other obstacles, causing debris that injures or causes fatalities on the ground.	Smart Return home is always active, thereby preventing aircraft from landing outside defined areas.
(e)	**RPAS loss of Control**	• Undesired flight trajectory could cause mid-air collision with other RPAS or manned aircraft. • Uncontrolled descent or landing could cause serious injury	Stabilisation and position holding based on GPS and INS reduces any uncontrolled action risk.

SER NO	EMERGENCY/ RISK/POTENTIAL HAZARD	POSSIBLE EFFECTS	CONTINGENCY PLAN/ FAILSAFE MEASURE
		to many persons on the ground and possible fatalities.	

CONCLUSION

11. Pre-flight operations are crucial for ensuring the safety and success of drone flights. By following a comprehensive checklist and adhering to safety guidelines, pilots can mitigate risks and conduct safe and responsible operations.

12. **Key pre-flight considerations include**:-

(a) **Aircraft Inspection**. Thoroughly inspect the drone for any damage or anomalies.

(b) **Airspace Assessment**. Verify airspace regulations, identify potential hazards, and obtain necessary authorisations.

(c) **Battery and System Check**. Ensure batteries are fully charged and the drone's systems are functioning correctly.

(d) **Pilot Proficiency**. Ensure the pilot is adequately trained and familiar with the drone's controls and emergency procedures.

(e) **Weather Conditions**. Assess weather conditions and avoid flying in adverse weather.

13. By prioritising safety and following established procedures, drone pilots can maximise the benefits of this technology while minimising potential risks.

- **Battery Check**. Ensure the drone's battery is fully charged and in good condition.
- **Propeller Inspection**. Check for any damage or wear on the propellers.
- **Motors**. Verify that all motors are functioning correctly.
- **GPS Signal**. Ensure the drone has a strong GPS signal.
- **Compass Calibration**. Calibrate the drone's compass to ensure accurate navigation.
- **Camera Settings**. Check the camera settings and functionality if your drone has one.

- **Firmware Updates**. Make sure the drone's firmware is up to date.
- Weather Conditions. Assess the weather conditions to ensure they are suitable for flying.
- **Flight Controls**. Test the flight controls to ensure they are responsive.
- Pre-flight Test Flight. Perform a short test flight to check everything is working properly.
- **Loss of Signal**. Activate RTH before flight.
- **Battery Emergencies**. Initiate RTH or safe landing.
- **GPS Signal Loss**. Use position hold mode for manual control.

HIGHER-ORDER THINKING SKILLS (HOTS)

➢ **Risk Assessment**. Analyse the importance of pre-flight checks in drone operations. How can neglecting these checks lead to potential risks, and what preventive measures can be put in place to mitigate these risks?

➢ **System Integration**. Evaluate the integration of automated systems in pre-flight checks. How can AI and machine learning enhance the accuracy and efficiency of these checks, and what challenges might arise from this integration?

➢ **Scenario Planning**. Imagine a situation where a drone experiences a critical system failure mid-flight. Develop a comprehensive emergency protocol that includes immediate actions, risk management, and post-incident analysis.

➢ **Ethical Implications**. Discuss the ethical considerations involved in drone operations, particularly in urban areas. How can pre-flight checks and operational protocols be designed to ensure public safety and privacy?

➢ **Technological Advancements**. Investigate the role of emerging technologies such as Internet of Things (IOT) and 5G in improving drone pre- flight checks and operations. What benefits do these technologies offer, and what potential drawbacks should be considered?

➢ **Environmental Impact**. Assess the environmental impact of drone operations. How can pre-flight checks be modified to ensure drones operate efficiently and minimise their ecological footprint?

ASSESSMENT EXERCISE

Multiple Choice Questions

Q1. What should you check for any signs of damage or swelling before a flight?

(a) Motors (b) Propellers
(c) Battery (d) GPS

Q2. Which component should be inspected for cracks, bends, or chips?

(a) Camera (b) Propellers
(c) Chassis (d) Controller

Q3. What should be calibrated in an open area to ensure accurate navigation?

(a) Camera (b) Battery
(c) Compass (d) Motors

Q4. Which update should you check for and install before a flight?

(a) Weather report (b) Firmware
(c) Software (d) Flight logs

Q5. What should you check for a strong signal before taking off?

(a) Wi-Fi (b) Bluetooth
(c) GPS (d) Radio

Q6. What would you perform to check the drone systems are functioning correctly before the actual flight?

(a) Full flight (b) Ground test
(c) Short test flight (d) Visual inspection

Q7. Which weather conditions should you avoid flying in?

(a) Mild breeze (b) Clear skies
(c) High winds and rain (d) Cloudy

Q8. What should you test for responsiveness to ensure full control of the drone?

(a) Motors (b) Flight controls
(c) Propellers (d) GPS

Q9. What must be in good condition to prevent mid-air power loss?

(a) Motors (b) Chassis
(c) Battery (d) Propellers

Q10. Which component needs to have its cleanliness checked to guarantee clear footage?

(a) GPS (b) Compass
(c) Camera lens (d) Firmware

Q11. What should be spun by hand to check for smooth operation?

(a) Propellers (b) Motors
(c) Compass (d) Battery

Q12. Which condition requires calibration to avoid flight issues?

(a) Camera settings (b) Battery levels
(c) Compass calibration (d) Propeller alignment

Q13. Which component's settings should be checked if the drone has one?

(a) Propellers (b) Camera
(c) Battery (d) Motors

Q14. What kind of low-altitude test should be conducted before a flight?

(a) Emergency landing test (b) Battery drain test
(c) Preflight test flight (d) Weather resistance test

Q15. What should be assessed to ensure they are suitable for flying?

(a) Wind speed (b) Weather conditions
(c) Propeller condition (d) GPS signal

Q16. Which control is responsible for changing the altitude of a drone?

(a) Roll (b) Pitch
(c) Yaw (d) Throttle

Q17. Which control adjusts the drone's forward and backward movement?

(a) Roll (b) Pitch
(c) Yaw (d) Throttle

Q18. What does the yaw control effect on a drone?

(a) Altitude
(b) Horizontal movement
(c) Rotation around the vertical axis
(d) Speed

Q19. Which control would you use to make the drone move left or right?

(a) Roll
(b) Pitch
(c) Yaw
(d) Throttle

Q20. Combining which two controls allows the drone to fly in a diagonal direction?

(a) Throttle and yaw
(b) Pitch and roll
(c) Roll and throttle
(d) Pitch and yaw

Q21. What is the first step you should take if your drone loses signal?

(a) Increase altitude
(b) Turn off the controller
(c) Initiate Return to Home (RTH)
(d) Fly in manual mode

Q22. What should you do if your drone starts drifting uncontrollably?

(a) Increase throttle
(b) Perform a compass calibration
(c) Reduce throttle and land immediately
(d) Switch to sport mode

Q23. If you notice a low battery warning in mid-flight, what action should you take?

(a) Fly further away to complete your mission.
(b) Ignore the warning.
(c) Return the drone to the home point as soon as possible.
(d) Land it in the nearest safe area.

Q24. In the event of a motor failure during flight, what is the best course of action?

(a) Try to regain control
(b) Perform an emergency landing
(c) Increase throttle
(d) Ignore the failure and continue flying.

Q25. What should you do if your drone encounters strong unexpected winds?

(a) Increase the speed to combat the wind.
(b) Fly lower and closer to the ground.
(c) Hover in place until the wind subsides.
(d) Ascend to a higher altitude.

Short Answer Questions

Q1. What should be ensured about the drone's battery before flight?

Q2. Why is propeller inspection important?

Q3. What needs to be verified regarding the drone's motors?

Q4. What should be checked to ensure accurate navigation?

Q5. How do you ensure the drone's camera is functioning properly?

Q6. Why are firmware updates important for drones?

Q7. What should be assessed before flying a drone?

Q8. Why is a pre-flight test flight important?

Q9. What should be activated before flight in case of a loss of signal?

Q10. What should be done during battery emergencies?

Long Answer Questions

Q1. Describe the steps involved in preparing a drone for flight, focusing on battery check, propeller inspection, and motor verification.

Q2. Explain the importance of GPS signal strength and compass calibration for drone navigation.

Q3. Discuss the procedures and checks needed to ensure the camera settings and firmware updates are up to date on a drone.

Q4. Analyse the factors to consider regarding weather conditions and flight controls before operating a drone.

Q5. Outline the emergency procedures to follow in the event of a loss of signal, battery emergencies, and GPS signal loss during drone flight.

ANSWER KEY : MULTIPLE CHOICE QUESTIONS (MCQ)